The Form of Dissent: Measuring Preference Structure in Multiparty Contexts

Abstract

Polarization is usually measured by mapping citizens onto a one-dimensional ideological scale and summarizing the resulting distribution. This paper instead treats dissent as a property of ranked-preference profiles. In multiparty settings, citizens rank several alternatives, and political conflict may appear not only as ideological distance, but as organization in the space of rankings.

I develop a structure-sensitive measurement strategy that separates global reversal patterns, effective support, and group-level structuration. Globally, Ψ, R, HHI, and $RHHI$ distinguish pairwise antagonism, reversal mass, and reversal concentration. Effective-number diagnostics track whether dissent is diffuse across many rankings or compressed into fewer effective configurations. At the group level, C, D, $1-O$, and S distinguish internal coherence, between-group divergence, distributional non-overlap, and consensus separation. I apply these diagnostics to ranked candidate preferences constructed from Brazilian CSES survey data from 2006, 2018, and 2022 using a nested resampling, imputation, and linearization pipeline.

The results show that Brazil's ranking profile becomes more conflictual and more organized between 2006 and 2022, but not as a collapse into two internally homogeneous camps. In 2006, dissent is present but relatively diffuse, with weak PT-centered structuration. In 2018, the profile becomes more conflictual and is organized most strongly by respondents' position toward Lula. By 2022, conflict is both higher and more compressed: pairwise antagonism and reversal mass rise, effective ranking and reversal-pair diversity fall, and ideology overtakes PT as the dominant structuring variable. The resulting pattern is compressed ideological structuration in a multiparty ranking space, not a simple binary division of the electorate. The results motivate a general point for comparative research: structure-aware metrics can register rising conflict without forcing electorates into binary representations, providing a portable template for dissent measurement in multiparty systems.

Keywords: political dissent; preference profiles; political polarization; multiparty systems; Brazil

1 Introduction

The global discourse on polarization is shaped by two facets: the widespread intuition that political conflict is both real and consequential, and the persistent ambiguity surrounding what polarization actually is. The World Economic Forum's *Global Risks Report 2025* lists polarization among the major short- to medium-term global concerns, alongside misinformation, inequality, and state-based armed conflict, emphasizing its perceived capacity to weaken trust, harden group boundaries, and amplify systemic risks (Elsner, Atkinson, and Zahidi 2025). Yet the concept remains overloaded. It is used to describe ideological distance, affective hostility, elite conflict, social sorting, partisan antagonism, and institutional deadlock. Even skeptical accounts, which question the extent or political relevance of polarization, often operate within the same conceptual ambiguity. The term carries implicit theoretical commitments about what kind of division matters, where it is located, and how it should be measured.

This ambiguity is especially consequential in multiparty settings. Much of the polarization literature relies on binary or one-dimensional representations, most commonly left versus right or government versus opposition. These representations are useful, but they foreground antagonism while often obscuring the organization of dissent itself. If citizens evaluate several alternatives, then political conflict is not exhausted by their location on a scale or by their first choice. It also appears in the structure of their relative evaluations: which alternatives are ranked together, which rankings are reversed, how concentrated those reversals are, and whether observable groups occupy distinct or overlapping regions of the ranking space.

Brazil offers a useful vantage point for this problem. It is frequently described as deeply divided in journalistic and political discourse, and recent elections have been marked by intense elite conflict and affective antagonism (Nunes and Traumann 2023; D. J. Samuels and Zucco 2018; Sen 2007). At the same time, Brazil's multiparty presidential system, fragmented electoral field, and heterogeneous electorate make it difficult to assume that a single ideological axis captures the relevant structure of political disagreement. The empirical question is therefore not only whether Brazil is polarized, but how dissent is organized in the space of ranked candidate preferences.

Dissent in multiparty electorates is not exhausted by attitudes, ideal points, or first choices. It also has a profile structure: a pattern in how citizens rank, reverse, cluster around, and separate among multiple alternatives. The contribution of the article is to develop diagnostics that make this structure observable and comparable. I develop diagnostics that describe how rankings are organized globally and across exogenous respondent partitions. Globally, Ψ, R, κ, $RHHI$, and effective-support summaries distinguish pairwise antagonism, reversal mass, reversal concentration, and the effective number of opposed rankings. At the group level, C, D, O, and S distinguish internal coherence, between-group divergence, distributional overlap, and consensus separation.

I then apply these diagnostics to ranked candidate preferences constructed from Brazilian CSES survey data from 2006, 2018, and 2022. The application shows why the distinction matters. Dissent increases and the ranking space becomes more compressed, but the resulting structure is not reducible to a binary split. In 2018, Lula support is the strongest structuring partition; in 2022, ideology overtakes PT. The analysis therefore describes not only whether conflict rises, but what form that conflict takes in profile space. I also separate variability induced by resampling, imputation, and stochastic linearization to assess the robustness of these descriptive patterns.

The scope of the paper is descriptive and measurement-oriented. In Gerring (2012)'s terms, the core contribution is a set of indicators: measurement claims that map observed rankings to scales in order to describe specific dimensions of an object, while clarifying what later explanatory or causal analyses would need to account for.[1] The broader aim is to contribute to the constitutive explanation of political conflict: the identification of its parts, internal organization, and observable forms (Dowding and Miller 2025). Rather than asking only whether an electorate is polarized, the paper asks what form dissent takes in profile space. This provides a portable measurement strategy for comparative research on multiparty electorates, where conflict may increase without becoming binary, and where group structuration may occur without complete within-group consolidation.

Section 2 reviews the literature on polarization measurement and argues that attitude-based and ideal-point approaches miss important features of ordinal preference structure. Section 3 defines the structure-sensitive diagnostics for ranked-preference profiles, including the global measures Ψ, R, κ, $RHHI$, effective-support summaries, and the group diagnostics C, D, O, and S. Section 4 describes the Brazilian CSES data, candidate-set construction, imputation, linearization, and uncertainty decomposition. Section 5 presents the empirical results for 2006, 2018, and 2022. The conclusion summarizes the argument and discusses how future work can extend these diagnostics to identify the candidate comparisons that generate internal dispersion or between-group divergence.

2 Why Study Polarization in Preferences?

The basic structure of the modal approach to polarization essentially consists of four elements: a set of agents, a set of attributes, a distribution over these attributes, and a measure defined on this distribution. Moreover, some notion of group, a partition of the set of agents, is also assumed (Bramson, Grim, Daniel J. Singer, et al. 2016; Esteban and Ray 1994). Analyses of polarization usually focus on two groups, contrasting their evaluations of two attitude objects (Fiorina and Abrams 2008; Mason 2015; Iyengar, Sood, and Lelkes 2012). Attitudes are absolute evaluative

[1] Thus, the paper does not estimate causal effects of institutions, campaigns, or elite rhetoric on preference structuration. Addressing those causal questions would require additional design, data, and assumptions beyond the present scope.

judgments directed at specific objects (Holbrook and Druckman 2011). In political contexts, such objects are typically candidates, parties, organizations, or social groups. The set of agents is either citizens, or politicians. The main underlying group identifier tends to be political parties. However, recent work on mass polarization has sought to go beyond this binary framing (Kekkonen and Ylä-Anttila 2021; Rothers 2025; Mehlhaff 2023), motivated by the fact that many countries exhibit an effective number of parties (ENP) greater than two. I propose by moving beyond the binary framework we should also extend the analysis to include relative evaluative judgments, that is, citizens' preferences. Why?

Person / Candidate	**C1**	**C2**	**C3**
P1	**100**	60	0
P2	60	**85**	70
P3	50	55	**60**
P4	52	56	**60**
P5	51	54	**59**
Mean	62.60	62.00	49.80

Table 1: Feeling-thermometer scores. Bold indicates each respondent's most preferred candidate.

First, attitude-based measures often summarize evaluations by averaging responses across individuals (Fuks and Marques 2022; Borba, Ribeiro, and Fuks 2024; Mehlhaff 2023). This operation is meaningful only under interpersonal comparability assumptions. To average thermometer scores is not merely to summarize responses; it is to treat numerical levels and intervals as comparable across persons. That assumption is substantive and usually unverified. Table 1 illustrates the issue. The mean scores rank the candidates as $C1 \succ C2 \succ C3$, but this result depends on treating P1's use of the full 0–100 scale and P3–P5's compressed use of the 50–60 band as measurements on the same interpersonal scale.

By contrast, the ordinal information in the same table requires only each respondent's internal ordering of the alternatives. It does not assume that a score of 60, or a difference of ten points, has the same meaning across respondents. At the ordinal level, three of the five respondents rank $C3$ first, and standard ordinal procedures such as pairwise majority comparison or Borda produce $C3 \succ C2 \succ C1$.[2] Using preferences thus avoids a comparability assumption that is not generally guaranteed in survey data (Narens 2007; Narens and Skyrms 2020; Sen 2018).

Second, while unidimensional Euclidean ideal-point methods are preference-based, they rely on a spatial representation that presuppose single-peakedness on the profile (Elkind, Lackner, and

[2]The point is not that the ordinal ordering is the uniquely correct social ordering, or that cardinal and ordinal summaries must coincide under full interpersonal comparability. They need not. The point is one of meaningfulness: cardinal aggregation across persons presupposes interpersonal comparability of levels and intervals, whereas ordinal profiles do not.

Peters 2022; Fiorina and Abrams 2008). Spatial models locate voters and alternatives on a line and define utility as decreasing in distance from each voter's ideal point. This construction implies that, along the relevant ordering of alternatives, each voter has a single peak and ranks alternatives lower as they move farther from that peak. Hence, if no ordering of the alternatives makes the observed profile single-peaked, a unidimensional Euclidean representation cannot preserve the individual rankings.[3]

This is not a marginal concern. Violations of single-peakedness in citizens' rankings over candidates is widespread on survey data (Regenwetter et al. 2006; Feld and Grofman 1988; Ballester and Haeringer 2010). In such cases, distances from putative ideal points do not recover the ordinal information contained in voters' preferences, and aggregation based on those points inherits the mismatch. The problem is muted in two-alternative settings, where single-peakedness is vacuous, but it becomes substantive as soon as voters rank three or more alternatives. This is precisely the setting in which multiparty electorates require methods that preserve more of the structure of preference profiles.

Table 2: Preference profiles with the same first choices but differing structure

Profile 1	**Profile 2**
ABCD	ADCB
ABDC	ADCB
ACBD	ADCB
ACDB	ADCB
ADCB	ADCB
ADBC	ADCB
BACD	BCDA
BADC	BCDA
BCAD	BCDA
BCDA	BCDA
BDAC	BCDA
BDCA	BCDA

Third, when there are more than two alternatives, important patterns, such as partial agreements or localized dissent, tend to be obscured. The resulting narrative becomes reductive and potentially misleading. Consider the preference profiles presented in Table 2. Each row represents the preference of a voter. For example, in row 1 of Profile 1, the voter strictly prefers A over B over C over D. If we assess polarization solely on the basis of voters' first choices, these two profiles appear identical and equally polarized. However, this interpretation does not withstand closer scrutiny,

[3]This claim concerns unidimensional Euclidean representations. Single-peakedness is not a necessary condition for embeddings in higher-dimensional spaces (Elkind, Lackner, and Peters 2022).

due to the substantial loss of information involved in reducing polarization to the distribution of first preferences. Such an approach completely ignores the underlying structure of support for alternatives C and D. Profile 1 exemplifies maximal diversity beyond the first choices, A and B. Every possible ranking involving the alternatives B, C, D and C, A, D occurs with equal probability in the subprofiles. That is, if we split Profile 1 into groups based on first choices, with one group preferring A and the other preferring B, we observe minimal internal coherence within each group.

Profile 2, by contrast, reveals a distinct structure. When divided by first choices, it displays maximal internal coherence, since each group is internally unanimous, and maximal intergroup divergence, because the rankings of voters who prefer A are completely reversed relative to those who prefer B. Remarkably, however, Profile 2 also reveals a consensus absent in Profile 1. The alternatives C and D appear more inclusive than A and B (Igersheim et al. 2022). The visible "polarization" therefore conceals an underlying consensus, as C and D are broadly acceptable.

This analytical example gains relevance when contrasted with narratives based on margins of victory. Consider the 2022 Brazilian presidential election, which Nunes and Traumann (2023) cites as blatant evidence of a deeply polarized country. Lula won the second round, under a majority-with-runoff system, with 50.9% of valid votes, compared to 49.1% for Bolsonaro, a margin of less than one percentage point. This was the closest victory in the history of Brazilian presidential elections. The second closest occurred in 2014, when Dilma won with 51.64% against 48.36% for Aécio Neves. Does this imply that the country was polarized? As the analytical example shows, any conclusion must be qualified by the rest of citizens' rankings.

Accordingly, there is a need for alternative measures that more faithfully capture the structure of preference profiles. Even within attitude-based approaches, polarization is a multidimensional concept (Bramson, Grim, Daniel J Singer, et al. 2017; Bramson, Grim, Daniel J. Singer, et al. 2016). It is therefore unlikely that any single measure can adequately capture it (Patty and Penn 2019; Patty and Penn 2014).

3 Global and Group Structure in Preference Profiles.

We now turn to the development of measures that can capture the internal structure of preference profiles. The challenge lies in designing indices that are sensitive not only to the distribution of first choices, the traditional focus of polarization research,but also to the patterns of agreement and disagreement that emerge across the full ranking of alternatives.

Can, Ozkes, and Storcken (2015) propose a measure of polarization that can be applied to profiles of linear orders[4], but it has not yet been empirically mobilized. Let $\mathcal{L}$ be the set of linear

[4] A *strict linear order* on a non-empty set X is a binary relation $<$ that is (i) irreflexive: $\neg(x < x)$ for all $x \in X$; (ii) transitive: $x < y \wedge y < z \implies x < z$; and (iii) trichotomous: for any distinct $x, y \in X$, exactly one of $x < y$ or $y < x$

orders on a finite set of alternatives A. Given a profile $p \in \mathcal{L}^N$ two distinct alternatives $a, b \in A$, we define $n_{ab}(p) = \#\{i \in N : (a, b) \in p(i)\}$, the number of individuals who prefer a over b; and $d_{ab}(p) = |n_{ab}(p) - n_{ba}(p)|$, the absolute difference in relative support between $a \succ b$ and $b \succ a$ in the population. If n is the number of agents, m is the number of alternatives, and $\bar{A}$ is the set of all subsets of A of size 2, then the measure is given by:

$$\Psi(p) = \sum_{\{a,b\} \in \bar{A}} \frac{n - d_{ab}(p)}{n \cdot \binom{m}{2}},$$

Despite its elegance, the measure Ψ has relevant limitations. It considers only global margins between pairs of alternatives—that is, the aggregate disagreement between pairs. It is not really using the preference information. Note that Ψ is just a summation of margins of victories, normalized by the number of pairs of alternatives. We could have a list of incomplete pairwise evaluative judgements ($x \succ_i y$, for not all alternatives), or even voters with cyclic preferences and the measure would be insensitive to these variations. As such, it completely ignores structural features of profiles, such as the formation of clusters of preferences (Bramson, Grim, Daniel J. Singer, et al. 2016). This insentivity to the structure of the profiles implies that disparate profiles with the same margins of victory will be considered equally polarized. To present an extreme and relevant case first let's note that the reversal of an ordering $abcd$ is $dcba$. Now, note that Ψ assigns the maximum value 1 both to: a uniform profile, in which all possible linear orders are equally probable (with probability $\frac{1}{m!}$), and to a profile composed of any two reversed preference orders with equal proportion of $\frac{1}{2}$.

One can remain agnostic as to which of the two profiles should be regarded as "more polarized." Analogous to what is argued in Section 2, the first profile exhibits maximal diversity, whereas the second is highly concentrated on a single opposition. In the latter, however, certain candidates remain relatively consensual, eliciting neither strong rejection nor strong approval. It is desirable to differentiate between these configurations, though, and Ψ fails to do so.

Moreover, recent literature has increasingly emphasized the explicitly group-based nature of the concept of polarization (Mehlhaff 2023; Bramson, Grim, Daniel J Singer, et al. 2017). Conceptually, a influential formulation of polarization[5] treats it as the joint occurrence of high intragroup coherence and high intergroup divergence. The measure Ψ, however, is defined globally for the entire profile and does not incorporate a group structure. It therefore cannot serve as an operationalization of polarization under this group-based conception.

The first step to distinguish between preference profiles is to actually operate with the rankings

holds.

[5]Bramson, Grim, Daniel J Singer, et al. (2017) and Bramson, Grim, Daniel J. Singer, et al. (2016) identify nine distinct senses of polarization; the specific combination of these senses adopted yields different conceptualizations, and hence different measures, of polarization.

rather than with the aggregated pairwise comparisons. Note that a natural idea of polarized preferences is the opposition between rankings and their reversals. From a micro perspective, reversed preferences are those that exhibit the maximum distance according to standard metrics (Zwicker 2016); they therefore represent the case of maximum disagreement between two agents. Thus, agents with reversed preferences would have the lowest probability of mutually influencing each other, *ceteris paribus* (Flache et al. 2017; Maciel and Martins 2020). Note also that, from a macro perspective, the more reversed preferences we add to a profile, the more we inflate the value of Ψ. The fraction of the profile composed of directly opposing orderings capture features of the profile that foster global disagreement and are guaranteed to increase Ψ. To calculate this fraction first define the degree of opposition, or balance, between an ordering r_i and its reversal $-r_i$ as:

$$\mathrm{op}(r_i) = 2 \cdot \min(p(r_i), p(-r_i)),$$

where p denotes the proportion of each ordering in the profile. The total opposition in the profile is then given by:

$$R = \sum \mathrm{op}(r_i),$$

summing over the unique orderings r_i, so as to avoid double counting of reversed pairs.

To incorporate the notion of concentration, the Herfindahl–Hirschman Index (HHI) is applied to the proportions of reversal. Thus, define $q_i = \frac{\mathrm{op}(r_i)}{R}$, and we have

$$\kappa = \sum_i q_i^2.$$

The final measure, RHHI, is defined as the geometric mean between the degree of opposition and its concentration:

$$\mathrm{RHHI} = \sqrt{R \times \mathrm{kappa}}.$$

Why a product, rather than, say, an arithmetic mean? Because the construct we want is explicitly *conjunctive*: it should be large only when two features co-occur, namely (i) a substantial mass of perfectly opposed preference pairs (R) and (ii) a strong concentration of that opposition around a small set of pairings (κ). Either ingredient on its own describes a different situation: high R with low κ is widespread but diffuse antagonism, while low R with high κ is concentrated but small-scale antagonism. An arithmetic mean would treat these components as partially interchangeable by allowing one to compensate for the other, thereby assigning high scores to profiles that lack one of the defining ingredients. The geometric mean enforces the intended "both at once" logic: it increases only when both inputs increase, collapses toward zero when either component is near zero, and penalizes imbalance while remaining symmetric in R and κ. An interesting measure that

can be derived from κ is the *effective number of pairs of reversed preferences* (ENRP), by analogy with the effective number of parties (Shugart and Taagepera 2017) and is defined as $ENRP = 1/\kappa$, which facilitates the interpretation of values of κ. Given this connection between $RHHI$ and the dispersion of the profile, I also consider the *effective number of observed rankings* (EO) as a diagnostic in the analysis.

The simplest way to implement a group-based measure of profile structuration, while preserving the full information of orderings, is to evaluate exogenous groupings such as race, sex, religion, ideology, or candidate support. For each group, I estimate a consensus ranking directly from the observed rankings rather than assuming a parametric ranking model. Let $\mathcal{L}(A)$ be the set of all linear orders over the active candidate set A. For a group g, with members $i \in g$ and observed rankings $\rho_i \in \mathcal{L}(A)$, define the total Kendall loss of a candidate consensus ranking $\rho \in \mathcal{L}(A)$ as

$$L_g(\rho) = \sum_{i \in g} d_\tau(\rho_i, \rho),$$

where d_τ is Kendall tau distance. The group consensus set is then

$$M_g = \arg\min_{\rho \in \mathcal{L}(A)} L_g(\rho).$$

Thus, the consensus is a ranking, or set of rankings, that minimizes the total distance to the group's observed preferences.

When M_g contains more than one ranking, I do not impose an arbitrary tie-breaking rule. Instead, group-level quantities that depend on the group consensus are averaged over all minimizing rankings in M_g. This preserves the ambiguity generated by the profile itself. Once the consensus set of each group is obtained, it becomes possible to measure intragroup coherence and intergroup divergence.

We start with the coherence of a group g, defined in terms of the average Kendall tau distance between the group members' rankings and the group's consensus set. The Kendall distance, d_τ, quantifies the disagreement between two orderings by counting the number of pairs of alternatives ordered differently (Zwicker 2016). For a group g of size n_g, the consensus-relative average distance is:

$$\text{AvgDist}_g = \frac{1}{n_g} \sum_{x \in g} \frac{d_\tau(x, \rho_g)}{\binom{m}{2}}.$$

The coherence of group g is then defined as:

$$C_g^\star = 1 - \text{AvgDist}_g.$$

Thus, $C_g^\star = 1$ if all members share exactly the ordering ρ_g, and it decreases as internal divergence increases. We aggregate these coherences into a global score weighted by group size:

$$C^\star = \frac{1}{n}\sum_{g=1}^{k} n_g\, C_g^\star, \quad \text{with } n = \sum_{g=1}^{k} n_g.$$

A key correction is that $C^\star$ is not bounded below by 0, but by $1/2$. Intuitively, because ρ_g is chosen to minimize average Kendall distance, for each pair of alternatives one can orient ρ_g to agree with at least half of the group on that pair; hence the fraction of pairwise disagreements cannot exceed $1/2$, so $\text{AvgDist}_g \leq 1/2$ and therefore $C_g^\star \geq 1/2$ (and likewise $C^\star \geq 1/2$).

Since $1/2$ functions as a natural "no-structure" baseline, we rescale coherence to place this baseline at zero:

$$C \;=\; 2C^\star - 1,$$

so that $C \in [0, 1]$, with $C = 0$ corresponding to coherence no better than the baseline and $C = 1$ corresponding to perfect within-group unanimity.

Let k be the number of identified groups and n_i the size of group G_i. Let ρ_j denote the consensus ordering of group j. We define the measure of divergence between groups D as:

$$D = \frac{1}{(k-1)}\sum_{i\neq j}\frac{n_i}{n}\cdot\left(\frac{1}{n_i}\sum_{x\in G_i}\frac{d_\tau(x,\rho_j)}{\binom{m}{2}}\right).$$

The expression can be understood as follows. The sum $\sum_{i\neq j}$ runs through all ordered pairs of distinct groups, so we divide by $(k-1)$ to ensure normalization. Each group is weighted by its relative size using the factor n_i/n. The distance term $d_\tau(x,\rho_j)$ represents how far an individual in group G_i is, on average, from the consensus ordering of another group G_j, normalized by the total number of pairwise comparisons. Finally, the inner mean, $\frac{1}{n_i}\sum_{x\in G_i}(\cdots)$, captures this average distance across all members of G_i relative to the consensus ordering of G_j. The value of D increases as the groups become more distinct from each other and equals zero if all preferences are essentially identical.

The measure D captures how far the members of one group are from the consensus rankings of other groups. For this reason, it is useful as a measure of external divergence, but it is not a pure between-group statistic. It measures the average distance from members of one group to the consensuses of other groups. If a group is internally dispersed, its members will already be far from their own consensus, and this dispersion carries over into their distances to other groups' consensuses. Thus, a high value of D can reflect either genuine separation between group centers or simply the fact that the group is internally spread out. This motivates a separate measure of support

separation. Since $C^\star$ is raw within-group coherence, $1 - C^\star$ is the corresponding within-group dispersion term. I define

$$S = D - (1 - C^\star).$$

Thus, S measures the excess distance from other groups' consensuses after removing the within-group dispersion already contained in D. Values near zero indicate that other groups' consensuses are not much farther from a group's members than the group's own consensus is. Larger values indicate that the exogenous partition creates additional separation in the ranking space.

Consensus-based distances, however, do not exhaust group structuration. Two groups may have distinct consensuses while still assigning substantial probability to the same rankings, or they may have similar consensuses while differing in the distribution of rankings around those consensuses. I therefore also use overlap, O, as a distributional check. Whereas C, D, and S summarize distances to consensus rankings, O compares the empirical ranking distributions of groups directly. High overlap indicates shared support in the same regions of the ranking space; low overlap, or high $1 - O$, indicates distributional separation. Let $p_g(r)$ denote the empirical share of group g whose observed ranking is $r \in \mathcal{L}(A)$. For two groups g and h, define their distributional overlap as

$$O_{gh} = \sum_{r \in \mathcal{L}(A)} \min\{p_g(r), p_h(r)\}.$$

Thus, $O_{gh} = 1$ when the two groups have identical ranking distributions and $O_{gh} = 0$ when their supports are disjoint. In the aggregate, O is the size-weighted average of pairwise group overlaps. In the empirical figures, I report $1 - O$ so that larger values indicate greater distributional separation.

For this reason, I do not combine the group diagnostics into a single polarization score. A composite measure would obscure the distinction between internal coherence, external divergence, distributional overlap, and consensus separation. Instead, I treat C, D, O, and S as a diagnostic panel. The goal is not to assign each partition one scalar value of polarization, but to describe how, and in what sense, an exogenous grouping structures the preference profile. In this sense, the paper extends the disambiguation strategy of Bramson, Grim, Daniel J Singer, et al. (2017) and Bramson, Grim, Daniel J. Singer, et al. (2016) from one-dimensional attitude distributions to ranked-preference profiles: rather than asking whether an electorate is polarized in general, the analysis asks which structural features of dissent are present, and how they vary across time and partitions.

3.1 Measures in Action

Let us now consider a list of paradigmatic profiles. We define the following generating functions:

- u: generates a profile in which all individuals share the same ordering over m alternatives.

Table 3: Glossary of measures

Measure	Interpretation
Ψ	Pairwise antagonism across alternatives.
R	Mass of exactly reversed ranking pairs.
κ_R	Concentration of reversal mass.
$RHHI$	Composite of reversal mass and reversal concentration.
$ENRP$	Effective number of reversal pairs.
EO	Effective number of observed rankings.
C	Baseline-corrected internal coherence of groups.
D	Distance from group members to other groups' consensuses.
S	Excess divergence after removing within-group dispersion.
O	Distributional overlap among group ranking distributions.

Represented by the proportion map $\{r_\star : 1\}$.[6]

- e: generates a profile in which all $m!$ orderings appear exactly once. Represented by $\{r_1 : \frac{1}{m!}, \ldots, r_{m!} : \frac{1}{m!}\}$.[7]
- i: generates a profile in which half the individuals have an ordering $r_\star$ and the other half its reversal $-r_\star$. Represented by $\{r_\star : \frac{1}{2},\ -r_\star : \frac{1}{2}\}$.[8]

For any number of alternatives m, we have:

$$\Psi(i(m)) = \Psi(e(m)) = R(e(m)) = RHHI(i(m)) = 1 \tag{1}$$

$$\Psi(u(m)) = RHHI(u(m)) = 0 \tag{2}$$

However, the κ term in $RHHI$ interacts with the number of alternatives in profiles $e(m)$. Let us see:

$$R(e(3)) = 1, \quad \kappa(\mathrm{q}(e(3))) = \tfrac{1}{3} \Rightarrow RHHI \approx 0.577 \tag{3}$$

$$R(e(4)) = 1, \quad \kappa(\mathrm{q}(e(4))) = \tfrac{1}{12} \Rightarrow RHHI \approx 0.289 \tag{4}$$

$$\vdots \tag{5}$$

$$R(e(8)) = 1, \quad \kappa(\mathrm{q}(e(8))) \approx 0 \Rightarrow RHHI \approx 0 \tag{6}$$

[6] u for **u**nanimity.
[7] e for **e**quiprobable.
[8] i for **i**nversion.

Therefore, $RHHI$ depends on the number of alternatives in uniformly distributed profiles $e(m)$, approaching zero as m grows[9] By recalling that $\frac{1}{\kappa}$ gives us the effective number of pairs of reversed preferences we have that for $e(m)$,

$$ENRP(e(m)) = \frac{1}{\kappa((e(m)))} = \frac{m!}{2},$$

the maximum possible value. Thus, $e(m)$ profiles correspond to maximal dispersion (or minimal concentration) in preferences, which is reflected in the reversal component.

In this way, $RHHI$ not only distinguishes between the profiles $i(m)$ and $e(m)$, but also provides information on the distance between the profile and the maximum effective number of possible reversed pairs. In other words, $\kappa((p))$ acts as a simple measure of the distance between the minimum (1) and the maximum $\big(m!/2\big)$ number of effective pairs of reversed preferences[10].

Regarding the behavior of the group diagnostics C, D, S, and O, it is useful to interpret them against two reference profiles. First, consider an e-profile, where every possible ranking appears with equal frequency within each group. In this case, group labels do not organize the profile. The best consensus ranking still mismatches group members on about half of all pairwise comparisons, so $C^\star \approx 1/2$ and the rebased coherence is $C \approx 0$. The divergence statistic D, however, need not be close to zero, because it compares group members to other groups' fitted consensuses; in an unstructured ranking space, these consensuses are still typically at nonzero Kendall distance from individual rankings. This is why D must be interpreted together with separation measures. Since $S = D - (1 - C^\star)$, the within-group dispersion term absorbs the generic distance induced by the lack of structure, yielding $S \approx 0$. Overlap also diagnoses the same situation: if all groups approximate the same uniform distribution over rankings, then $O \approx 1$, or equivalently $1 - O \approx 0$.

Second, consider a unanimous profile u, where all respondents share the same ranking. Then internal coherence is maximal, so $C = 1$. If all groups share that same ranking, their consensuses coincide, $D = 0$, and therefore $S = 0$: there is no group separation because there is no disagreement to organize. Overlap is also maximal, $O = 1$. By contrast, if each group is internally unanimous but different groups hold different, possibly opposed, rankings, then $C = 1$, D is high, $S = D$, and $O = 0$. These reference cases show why the diagnostics should not be collapsed into a single score. C captures internal consolidation, D captures distance to external consensuses, S removes the within-group dispersion contained in D, and O checks whether groups actually share support in the ranking distribution.

[9] I explain that relationship on an appendix.

[10] Compare with the profile distance measure proposed by Szufa (2024), which involves solving optimization problems and is computationally costly.

4 Empirical Evidence from Brazil (ESEB 2006–2018–2022)

We now turn to the empirical application, drawing on data from the Estudo Eleitoral Brasileiro (ESEB) - which is associated with the Comparative Study of Electoral Systems for the years 2006 (n = 1000), 2018 (n = 2506), and 2022 (n = 2001). Brazil's multiparty presidential system has been invoked as a paradigmatic case of polarization, with its combination high electoral fragmentation, weak programmatic structuring, and the salience of a single polarizing party, the Workers' Party (PT), as a stable focal point of partisan affect (D. J. Samuels and Zucco 2018; Nunes and Traumann 2023; Elsner, Atkinson, and Zahidi 2025). The selected years correspond to qualitatively different political contexts as described in existing accounts of Brazilian competition and alignment.

The 2006 election took place during a period generally interpreted as one of relatively low mass polarization, despite growing elite antagonism following the mensalão corruption scandal. This contrasts sharply with 2018, widely recognized as a watershed moment of intense partisan and ideological polarization, marked by Jair Bolsonaro's rise and the consolidation of a strongly anti-PT bloc (D. Samuels, Mello, and Zucco 2023). The 2022 contest supposedly further entrenched this bipolar dynamic, producing the most competitive and contentious presidential race since redemocratization, with polarization manifesting both in elite rhetoric and in voter behavior (Nunes and Traumann 2023). Comparing these three waves allows a descriptive assessment of how preference-profile structure differs across periods commonly discussed in polarization narratives.

The analysis is based mainly on the following question present in the questionnaires:

- On a scale from 0 to 10, could you tell me what you think about the following political leaders? Remember that 0 means you do not like them at all, and 10 means you like them a lot. If you do not know the person or do not know how to answer, please say so.
  - 96. Does not know the politician
  - 97. Does not know
  - 98. Did not answer

This question is a thermometer scale and can therefore be interpreted as a weak order(Regenwetter et al. 2006)[11]. The list of political figures includes both declared presidential candidates and individuals publicly floated as potential contenders for the office.

Responses coded as 96, 97, and 98 are treated as missing values. Although there are minor coding variations between the datasets for the three years, the procedure is analogous. I consider

[11] A complete, reflexive, and transitive relation; in other words, an ordering that allows for indifference between alternatives.

profiles with orderings among 2 to 7 alternatives, either forcing the inclusion of certain substantively relevant sets or adding candidates based on the percentage of non-responses.

The starting point for the analysis is the construction of weighted pseudo-profiles, following the resampling strategy proposed by Popov, Popova, and Regenwetter (2014) and used in empirical social choice applications (Kaminski 2015; Lachat and Laslier 2024; Maciel 2024). In each replicate, respondents are resampled with replacement using the survey weights, producing a pseudo-profile on which the subsequent stages of the analysis are applied.

The design has three stochastic layers. First, weighted bootstrap resampling captures the variability induced by drawing pseudo-profiles from the observed survey. Second, missing candidate evaluations are imputed. The canonical specification uses MICE (Zhang 2016), while zero imputation is retained as a robustness specification. Third, the resulting weak preference information is linearized because the measures used below are defined on complete linear orders. I use pattern-conditional linearization as the canonical specification. Alternatives are first grouped by their observed rating levels, and rating levels are ordered from highest to lowest. For each indifference class, the procedure estimates a conditional distribution over compatible refinements using empirical tie-breaking patterns among respondents with comparable weak-order patterns. This distribution is regularized by adding one-half pseudo-count to each compatible refinement. Thus, well-populated weak-order strata are linearized mostly according to their empirical refinement frequencies, while empty or sparse strata are shrunk toward uniform tie-breaking. A simpler random-tie linearizer, which uniformly shuffles alternatives within each rating class, is used as a robustness specification.

This structure makes explicit that the measured profile is not observed directly, but obtained through a sequence of stochastic operations applied to the survey data. In the main analysis, I report one canonical specification of this sequence: weighted resampling, MICE imputation, and pattern-conditional linearization. I use this specification because it preserves more information from the survey than the alternatives: MICE uses the observed response structure to impute missing evaluations, and pattern-conditional linearization uses regularized empirical tie-breaking patterns rather than treating all indifferences as uniformly random. Measures are computed separately for each realized pseudo-profile generated under this specification, and the figures summarize the resulting distribution of replicate-level estimates. The uncertainty bands therefore reflect variation induced by the repeated construction of pseudo-profiles under the canonical pipeline. In the appendix, I retain the nested replicate indices and decompose this variability through stage-specific conditional variances associated with resampling, imputation, and linearization. The robustness checks show that the canonical specification tends to produce slightly higher measure values than the less informative specifications[12], and that the dominant source of simulation variability is

[12] show that

resampling rather than imputation or linearization.

The construction of the preference profiles requires fixing a set of evaluated political figures for each election year. This choice is not purely mechanical. Although response availability is an important constraint, selecting alternatives only by the lowest share of missing evaluations can confound electoral relevance with public notoriety. This is especially consequential in post-electoral surveys, where highly salient political figures may be widely recognized even when they were not relevant presidential alternatives in the election under analysis. I therefore use one substantively defined candidate sequence for each year[13]. For each year, profiles for smaller m are obtained by taking the first m alternatives from that sequence.

The main analysis is reported for $m \leq 5$. This cap is substantively and statistically motivated: since the number of possible rankings grows as $m!$, moving to $m = 6$ or $m = 7$ creates severe sparsity in ranking space[14].. Because the paper studies the structure of empirical preference profiles, this sparsity directly affects reversal mass, overlap, and concentration measures.

For each linearized pseudo-profile, I compute the global measures Ψ, R, HHI, and $RHHI$. I then compute the group-level measures C, D, and S after partitioning respondents by sex, race, religion, age, education, income, position regarding the Workers' Party (*Partido dos Trabalhadores*, PT), ideology, and opinion on abortion. Group consensuses are computed by minimizing average Kendall distance over the relevant catalog of linear orders, with consensus ties handled by averaging across tied minimizers. Measures are computed separately for each realized pseudo-profile generated by the nested design. The reported summaries are then computed from the distribution of replicate-level estimates, while the replicate indices are retained for the variance decomposition reported in the appendix. The recodings, candidate sequences, and scripts are available in the replication repository.

Elaborating on the choice of groups, I group respondents by standard sociodemographics (sex, race, religion, age, education, income) and by two political identifiers (ideology and PT orientation)[15]. The demographic cleavages can generate intragroup coherence via shared social positions, networks, and information environments (e.g., gender and cohort effects; racialized experiences; religion and church-based mobilization; education and income shaping media diets), and intergroup divergence when those contexts differ systematically. Ideology is self-identification on a left–right scale; even if potentially noisy at the individual level, in multiparty systems it might aggregate meaningful programmatic sorting. Position on abortion is included to check whether Brazil exhibits a cultural-moral cleavage akin to the U.S. case identified by DiMaggio, Evans, and Bryson (1996)[16]. Finally, Brazil's party system is often characterized by weak partisan attachments

[13] I discuss the selection of candidates in an appendix.

[14] I should actually test that

[15] I don't group the data by more than one group simultaneously.

[16] For a contrasting perspective, see Mouw and Sobel (2001), who also highlight the issue of interpretive validity in

and low ideological coherence (Borba, Ribeiro, and Fuks 2024), with a notable exception: the Workers' Party (PT), which elicits both strong support and strong rejection. Following D. J. Samuels and Zucco (2018), I include PT orientation to assess whether this party-centered cleavage, petismo vs. antipetismo, is reflected not just in attitudes but in the structure of preference dissent over political figures publicly mooted as presidential contenders. In short, these groupings are chosen because, if any observable partitions generate high C (within-group coherence) and high D (between-group divergence), these are the most theoretically plausible candidates in the Brazilian context.

4.1 Results

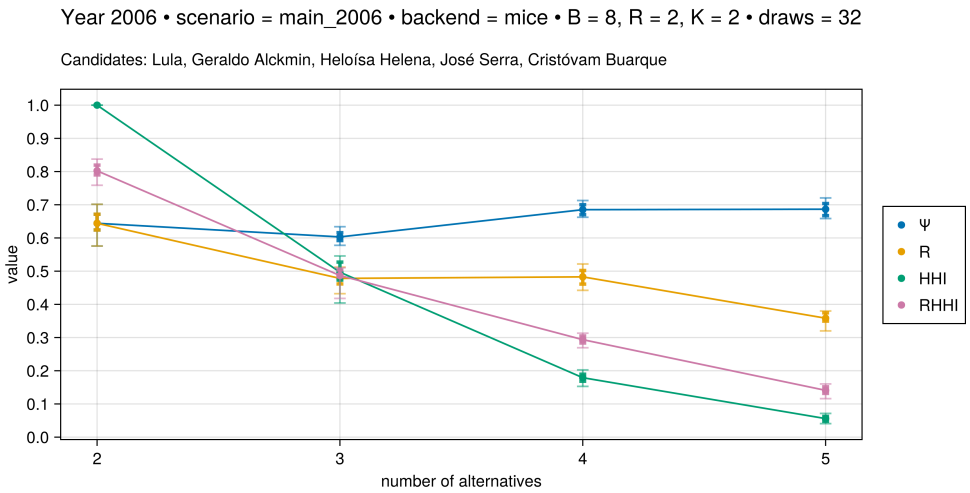
Figure 1 presents the median values[17] of the global measures for the candidate sets analyzed in 2006, 2018, and 2022, varying the number of alternatives m. The candidates appear in the order in which they were included in each scenario. The first mechanical pattern is that $\kappa = 1$ whenever $m = 2$. With only two alternatives, there is only one possible reversal field, $A \succ B$ versus $B \succ A$. Hence all reversal mass is necessarily concentrated in that field. For the same reason, Ψ and R coincide when $m = 2$, as shown in Appendix A.3. Since $RHHI = \sqrt{R \cdot \kappa}$, it follows that $RHHI = \sqrt{R}$ in the binary case and therefore exceeds $\Psi = R$ whenever $0 < R < 1$. Once $m \geq 3$, however, the measures separate because reversal can be distributed across multiple opposed ranking pairs.

The first substantive pattern is the temporal increase in pairwise antagonism. In 2006, Ψ is already moderate to high, beginning around 0.64 for the Lula–Alckmin comparison, falling slightly when Heloísa Helena is added, and then rising to approximately 0.68 or 0.69 when José Serra and Cristovam Buarque enter the candidate set. In 2018, Ψ is higher at every comparable value of m: it begins around 0.74 for Haddad and Bolsonaro and rises to approximately 0.81 for $m = 4$ and $m = 5$. In 2022, Ψ is higher still. It is around 0.92 from $m = 2$ through $m = 4$, declining only slightly to about 0.89 at $m = 5$. Thus, according to Ψ, pairwise antagonism increases sharply across the three waves, and by 2022 it remains high even after moving beyond the binary Lula–Bolsonaro comparison.

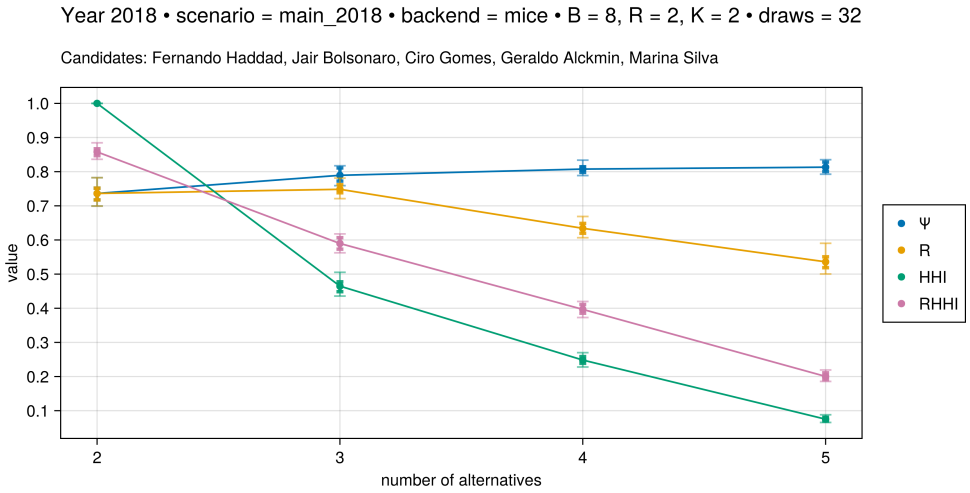
However, Ψ alone does not describe the structure of the profile. It summarizes average pairwise opposition, but it does not say whether the underlying ranking profile is organized around a small number of mutually reversed rankings or dispersed across many ranking configurations. This is why R, κ, and $RHHI$ have to be read together. The measure R captures the fraction of the profile involved in exact reversal pairs; κ captures how concentrated that reversal mass is across reversal fields; and $RHHI$ combines the two. A profile can therefore have high Ψ but lower $RHHI$ if

thermometer scores that I discuss in Section 2.

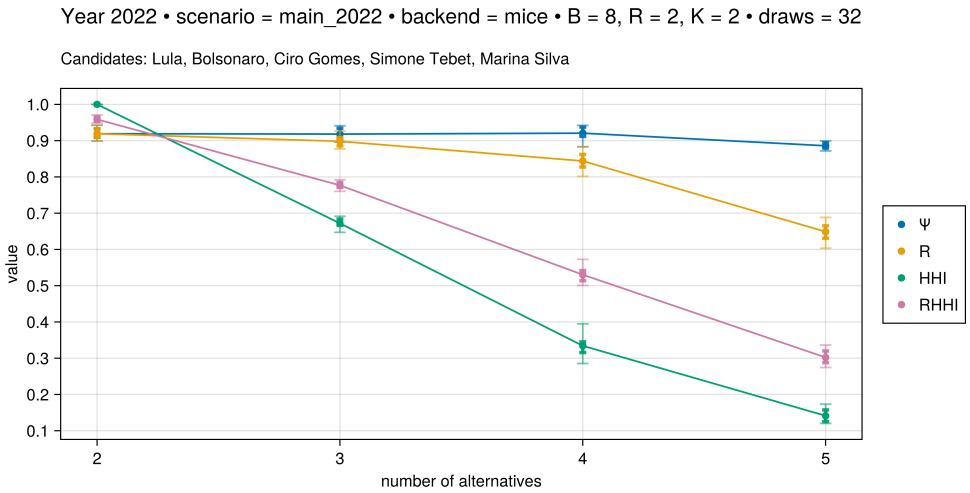
[17] All global measure plots show median values with quartile bands, while the group measure plots show medians.



(a) 2006



(b) 2018



(c) 2022

Figure 1: Evolution of global measures (2006, 2018, 2022)

pairwise antagonism is distributed across many rankings rather than concentrated in a few opposed rankings.

In 2006, this distinction is especially visible. Exact reversal mass R is about 0.64 at $m = 2$, falls to about 0.48 at $m = 3$, remains near 0.48 at $m = 4$, and then declines to roughly 0.36 at $m = 5$. The concentration component falls even more sharply: HHI drops from 1 at $m = 2$ to about 0.52 at $m = 3$, about 0.18 at $m = 4$, and about 0.05 at $m = 5$. As a result, $RHHI$ declines from about 0.80 in the binary comparison to about 0.49, 0.29, and 0.14 as the candidate set expands. The interpretation is not that conflict disappears in 2006. Rather, as the profile is represented over a richer ranking space, exact reversal mass becomes increasingly diffuse.

The 2018 profile is more conflictual than the 2006 profile. R is about 0.74 at $m = 2$, remains around 0.75 at $m = 3$, and then declines to approximately 0.64 and 0.54 at $m = 4$ and $m = 5$. These values are consistently above the 2006 values at comparable candidate-set sizes. The concentration component again falls with m, from 1 at $m = 2$ to roughly 0.47, 0.25, and 0.08. Consequently, $RHHI$ falls from about 0.86 to 0.59, 0.39, and 0.20. The important point is that the decline in $RHHI$ with m is compatible with a more conflictual profile. Compared with 2006, 2018 has higher pairwise antagonism, higher exact reversal mass, and higher reversal-concentration composites, but the opposition is still distributed over multiple reversal fields once more than two candidates are considered.

The 2022 profile displays the strongest global pattern. Exact reversal mass is about 0.92 at $m = 2$, remains near 0.90 at $m = 3$, is still around 0.84 at $m = 4$, and only then declines to approximately 0.65 at $m = 5$. This is a major difference from the earlier waves: reversal mass remains high even after the candidate set moves beyond the two leading candidates. κ still declines mechanically as the ranking space expands, from 1 at $m = 2$ to about 0.67, 0.33, and 0.14. But because R remains high, the composite $RHHI$ also remains much higher than in the previous years: approximately 0.96, 0.78, 0.53, and 0.30 from $m = 2$ to $m = 5$. In profile-space terms, 2022 combines very high pairwise antagonism with substantial exact reversal mass and comparatively stronger concentration of that reversal mass.

The effective number of reversal pairs, $1/\kappa$, makes the same point in more interpretable terms. With three alternatives, the theoretical maximum number of reversal pairs is 3. The effective number is about 1.9 in 2006, 2.1 in 2018, and 1.5 in 2022. Thus, at $m = 3$, 2022 has the largest reversal mass but the smallest effective number of reversal fields. Its opposition is therefore both stronger and more concentrated. With four alternatives, the theoretical maximum is 12, while the effective numbers are approximately 5.6, 4.0, and 3.0 in 2006, 2018, and 2022. With five alternatives, the theoretical maximum is 60, while the effective numbers are roughly 18, 13, and 7. The expansion of the candidate set necessarily opens many possible reversal fields, but the 2022 profile uses fewer of them effectively.

This decomposition clarifies why Ψ and $RHHI$ should not be interpreted as competing measures of the same object. Ψ captures the intensity of pairwise antagonism. $RHHI$ captures reversal mass weighted by its concentration across exact opposite rankings. The fact that Ψ remains high while $RHHI$ declines with m means that pairwise antagonism can be intense even when exact reversal is distributed across a large ranking space. Conversely, the fact that 2022 has higher $RHHI$ than 2006 and 2018 at comparable values of m means that its antagonism is not merely a marginal pairwise phenomenon. It is also more organized in the profile space.

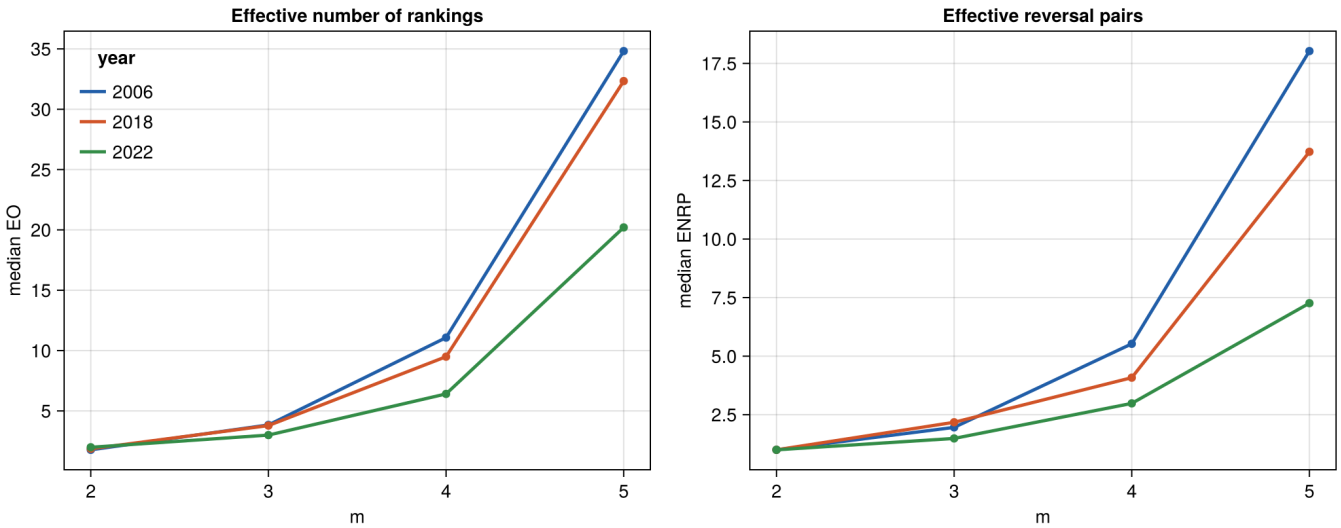


Figure 2: Compression of profiles according to EO and ENRP

After the global indices, Figure 2 adds a diagnostic view of the structure of the observed profile. The left panel reports the effective number of rankings, while the right panel reports the effective number of reversal pairs. Together, they show how much of the ranking space is effectively occupied and how diffuse or concentrated the exact-reversal structure is.

The first result is that the effective support naturally expands sharply with m, but not uniformly across years. At $m = 2$, the three elections are nearly indistinguishable: the profile is close to the two-ranking limit, and there is only one possible reversal pair. As m increases, however, the years separate. In 2006 and 2018, the effective number of rankings rises steeply, reaching approximately 35 and 32 effective rankings at $m = 5$. In 2022, by contrast, the profile remains much more compressed, reaching only about 20 effective rankings. Thus, although the formal ranking space grows factorially with m, the observed 2022 profile occupies a smaller effective subspace.

The reversal-pair diagnostic sharpens this contrast. By $m = 5$, 2006 has roughly 18 effective reversal pairs and 2018 has roughly 14, while 2022 has only about 7. This indicates that the exact-reversal mass in 2022 is more concentrated: opposition is organized through fewer structurally opposed ranking pairs. In 2006 and 2018, reversal mass is more diffuse, spread across a larger set

of opposing rankings[18].

Taken together, the global indices and the effective-support diagnostics suggest that the observed process is not simply a movement from pluralism to two-bloc polarization. Later waves show greater organization of disagreement, especially in 2022, where the ranking space is more compressed and reversal mass is more concentrated. Yet this concentration does not amount to dominance by a single antagonistic reversal pair.

When the choice set is binary ($m = 2$), the global measures partially collapse to the normalized pair margin. In this case, a close race, understood as a small aggregate margin between the two alternatives, is equivalent to high Ψ and high R. But the same binary restriction also forces $\kappa = 1$ and $ENRP = 1$. Hence, any further structure is unidentifiable: all opposition is concentrated in a single reversal field by construction. For this reason, reading a narrow electoral margin as "high polarization," as in Nunes and Traumann (2023, pp. 17–18), is misleading. It confuses a binary margin with the structure of the preference profile.

The point becomes visible as soon as $m \geq 3$. Once additional alternatives are admitted, reversal mass can be distributed across multiple opposed rankings. $RHHI$ therefore falls relative to the binary case, and $ENRP > 1$ shows that antagonism does not collapse onto a single divide. At the same time, the updated results show that this dispersion is not random fragmentation. In 2022, pairwise antagonism and reversal mass remain high even for larger candidate sets, while the effective number of reversal fields is smaller than in the earlier waves. The substantive conclusion is therefore more precise: binary margins overstate what can be inferred about polarization, but the multiparty profile reveals a real increase in structured opposition, especially in 2022.

Figure 3 reports the group-based diagnostics for 2006: within-group coherence C, between-group divergence D, distributional non-overlap $1 - O$, and consensus separation S. The main result is weak group structuration overall, with PT evaluation as the partial exception. For most demographic partitions, C is positive but modest, usually between about 0.32 and 0.41, and it declines or remains flat as the candidate set expands. Their D values are also moderate, but rarely distinctive: sex, ideology, and age are especially similar, while education and income are slightly higher. Thus, these partitions identify some differences in group consensuses, but they do not organize the profile strongly.

PT evaluation stands out in 2006. It has the highest coherence at $m = 2$, with $C \approx 0.47$, and the highest divergence, with $D \approx 0.53$. It also has the largest distributional non-overlap, rising from $1 - O \approx 0.39$ at $m = 2$ to $1 - O \approx 0.52$ at $m = 5$. The consensus separation measure S is

[18]The appendix table reports the same diagnostics numerically. It confirms that the separation between 2022 and the earlier elections becomes larger as m increases. The difference is especially pronounced for effective reversal pairs: at $m = 5$, 2022 has less than half the effective reversal-pair support of 2006 and only about half that of 2018. The substantive implication is that 2022 is not merely a point on a higher-or-lower polarization scale. It is a structurally more compressed profile, with conflict concentrated in fewer regions of the ranking space.

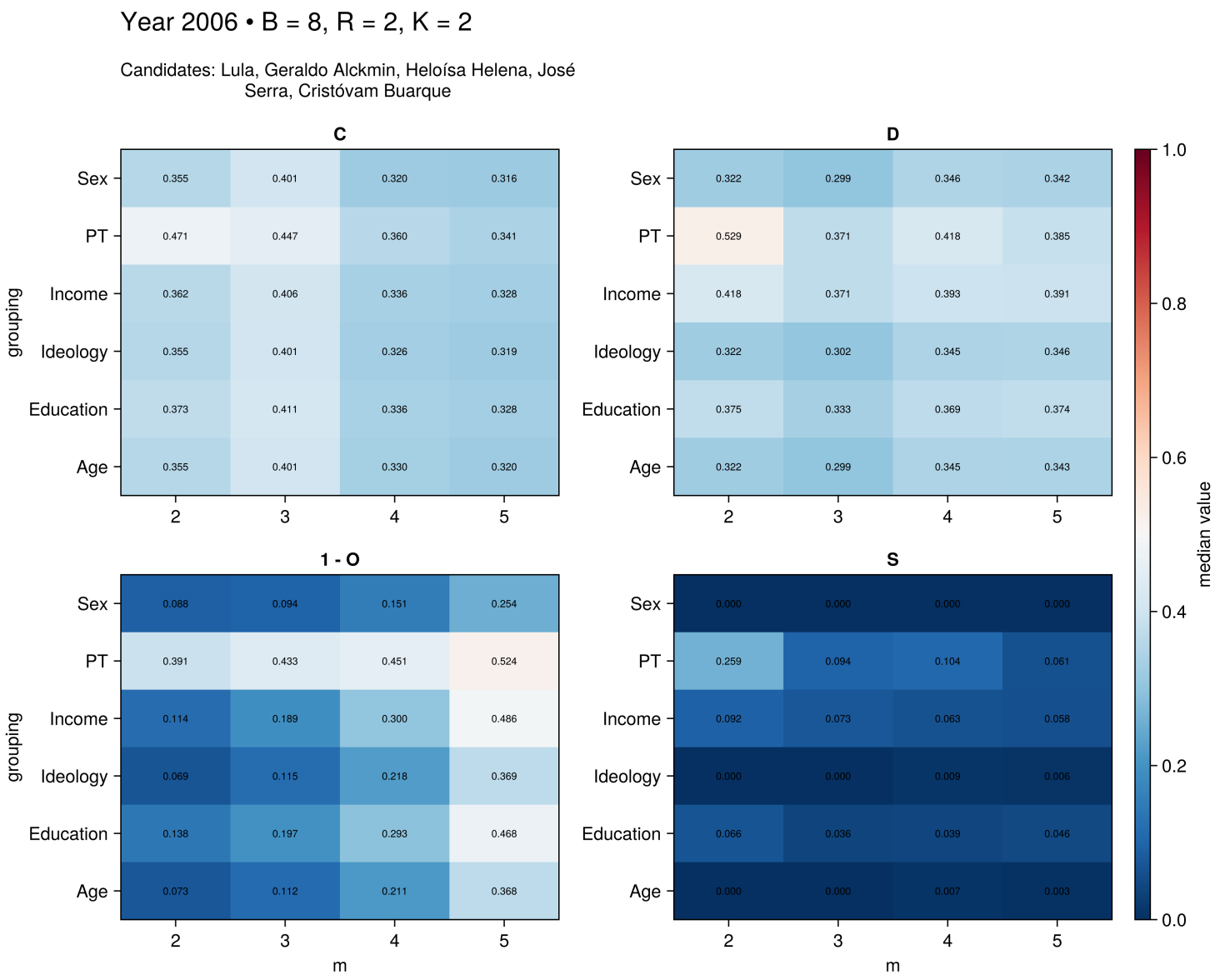


Figure 3: Group-based measures — 2006

the clearest indication of the difference: PT reaches $S \approx 0.26$ at $m = 2$, while sex, ideology, and age are essentially zero. However, this PT-centered structuration weakens as additional candidates enter the ranking set. By $m = 5$, PT's C has fallen to about 0.34, D to about 0.39, and S to about 0.06. Thus, in 2006, PT evaluation structures the binary comparison, but much less of the richer ranking profile.

The other 2006 groupings mainly show the distinction between non-overlap and consensus separation. For income, education, ideology, and age, $1-O$ increases with m, in some cases reaching values around 0.37 to 0.49. But this does not by itself imply strong polarization between groups. As m grows, the ranking space becomes larger and empirical support becomes sparser, mechanically reducing overlap across group distributions. The S panel is therefore more discriminating. It shows that most demographic partitions have little net separation once within-group dispersion is taken into account. In short, the 2006 profile contains global dissent and some PT-centered structure, but the observed demographic partitions do not generate compact and separated blocs.

Figure 4 shows a different pattern. The PT evaluation variable is not used because of the very high number of missing responses in 2018.[19] Instead, the relevant political grouping is respondents' score for Lula. This variable dominates the 2018 group diagnostics. It has the highest within-group coherence, beginning at $C \approx 0.51$ for $m = 2$ and declining to $C \approx 0.36$ by $m = 5$. It also has the highest between-group divergence, from $D \approx 0.56$ at $m = 2$ to $D \approx 0.53$ at $m = 5$. Its distributional non-overlap is high throughout, increasing from $1 - O \approx 0.43$ to $1 - O \approx 0.53$, and its consensus separation is the largest in the figure, declining from $S \approx 0.32$ to $S \approx 0.21$.

Among the groupings examined in 2018, ideology is the second most informative political partition. It is consistently above the demographic baselines, with C around 0.34 at $m = 2$ and 0.24 at $m = 5$, D around 0.47, and S declining from about 0.14 to 0.08. Religion also has relatively high D, near 0.45 to 0.47, and increasing $1-O$, but its S remains modest, around 0.09 at $m = 2$ and 0.06 at $m = 5$. This distinction matters: several groupings have differentiated consensuses, but only Lula's score combines high coherence, high divergence, high non-overlap, and high net consensus separation. The 2018 profile is therefore not simply more conflictual than 2006; among the tested partitions, Lula score is the clearest grouping associated with coherent and separated subprofiles, while ideology is weaker but still informative.

Compared with 2006, the 2018 demographic partitions show lower coherence and higher divergence. Sex, race, income, education, and age all have relatively low C, mostly near 0.20 to 0.28, while their D values often lie around 0.39 to 0.43. This means that these groups are internally less cohesive than comparable partitions in 2006, but their fitted consensuses are somewhat farther apart. The combined implication is not strong group polarization, because their S values remain small. The profile is more antagonistic globally, but demographic groups do not carry that antagonism in

[19] Approximately 1500 out of 2500 responses are missing.

a coherent way.

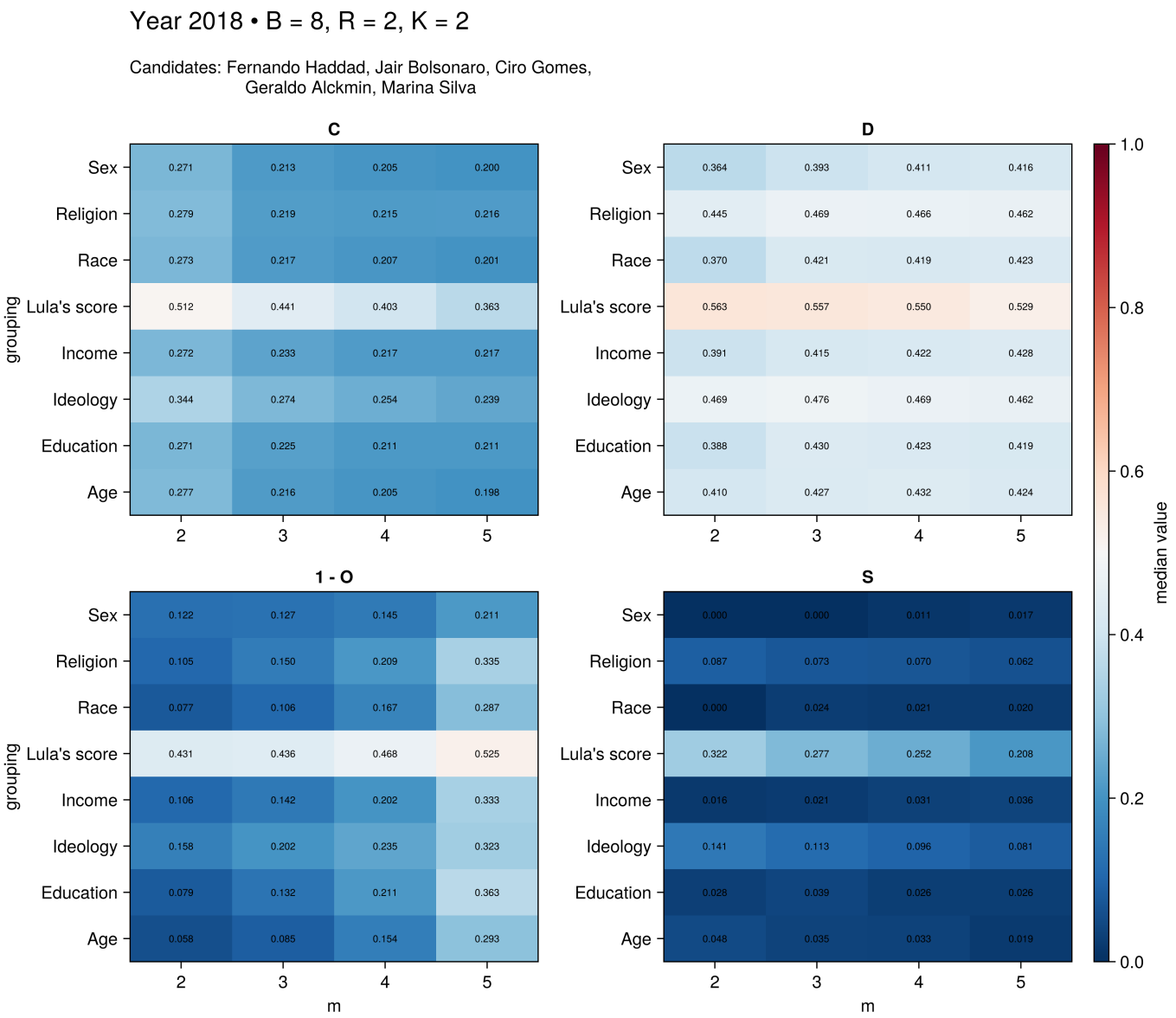


Figure 4: Group-based measures in 2018

In 2022, the group-based measures show a clearer structure than in the previous waves, but not because all groupings become equally informative. Race, age, education, income, and sex remain weak discriminators. Their within-group coherence C is low, their between-group divergence D is moderate, and their consensus separation S remains close to zero or only weakly positive across candidate-set sizes. These variables therefore do not define compact and externally separated preference blocs, even when their distributions become less overlapping as m increases.

The distinction between $1 - O$ and S is important here. For several demographic variables,

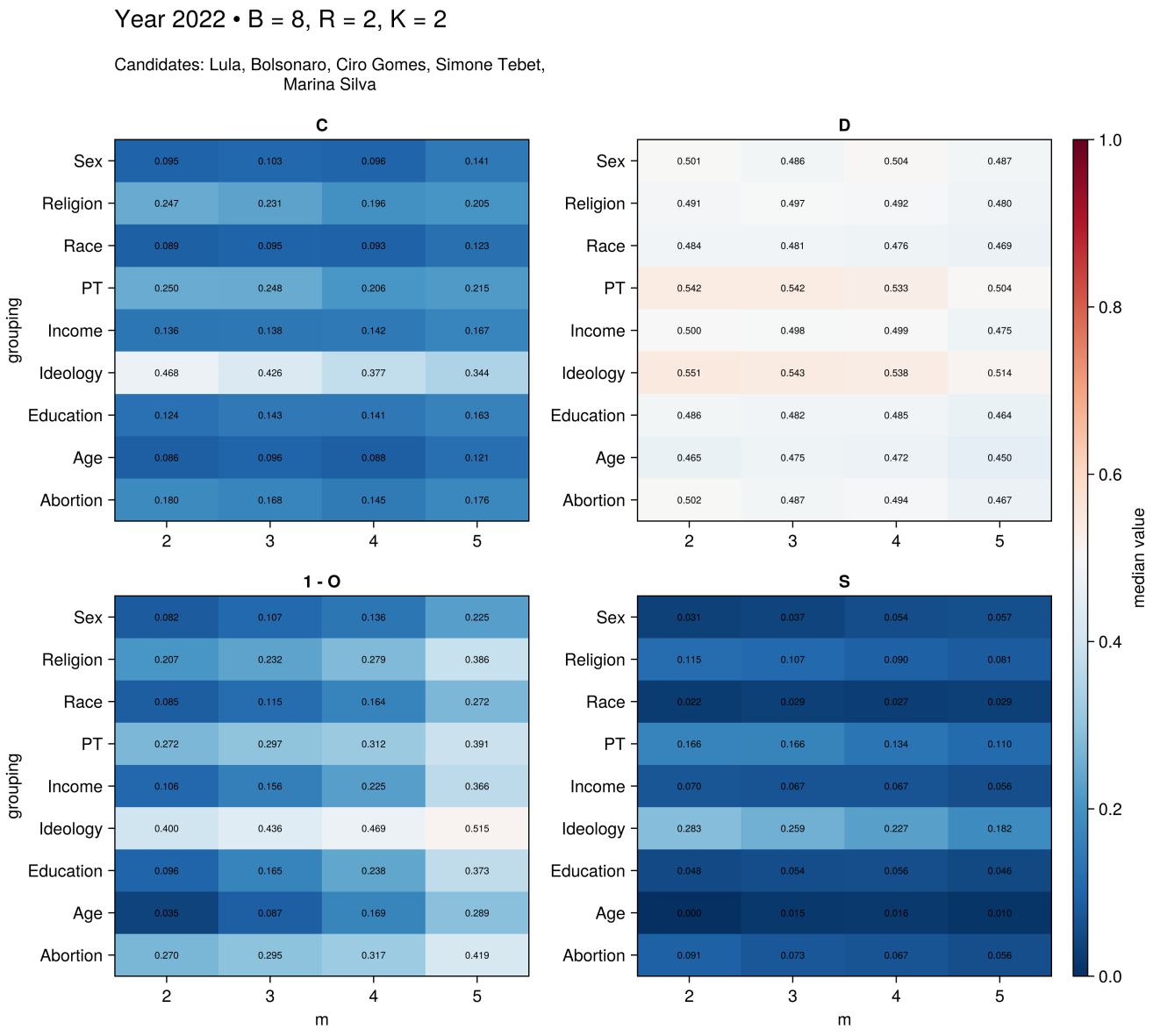


Figure 5: Group-based measures in 2022

$1 - O$ increases as alternatives are added. This does not by itself imply stronger polarization. As m grows, the ranking space expands, and empirical group distributions naturally become more sparse. A rise in non-overlap can therefore reflect the combinatorial expansion of the support rather than substantive group separation. The S measure is more restrictive: it asks whether between-group separation exceeds within-group dispersion. On this criterion, most demographic groupings remain weak.

Abortion, religion, and PT evaluation are more informative than the purely demographic partitions. Abortion shows moderate non-overlap and modest positive consensus separation, especially for smaller candidate sets, but it declines as new alternatives are included. Religion has relatively high divergence and non-overlap, but its coherence and S remain below the main political variables. PT evaluation is stronger: its D remains high across candidate-set sizes, indicating that PT-defined groups have distinct fitted consensuses. However, its within-group coherence is only moderate to low, and its S is positive but not dominant. Thus, PT evaluation still distinguishes group consensuses, but its within-group coherence is not high enough for it to be the most informative partition in 2022.

This marks a change from 2006. In 2006, PT evaluation was the main political grouping, especially in the binary comparison. By 2022, PT remains relevant, but its structuring role is weaker because the coherence component is lower. The decline is not driven primarily by a disappearance of between-group divergence: PT-defined groups remain far apart in consensus space. The more important change is that respondents within the same PT category are less internally coherent. In group-structuration terms, PT still differentiates groups externally, but it no longer produces the most internally organized subprofiles.

Ideology is the clear outlier in 2022. It has the highest within-group coherence C, the highest or near-highest between-group divergence D, the largest non-overlap $1 - O$, and the largest consensus separation S across candidate-set sizes. Its values also decline as more alternatives are added, which is expected: larger candidate sets make rankings harder to compress into internally coherent group consensuses. But even after this decline, ideology remains above the other partitions on the relevant diagnostics. Unlike PT, ideology combines internal coherence with external separation.

The substantive conclusion is therefore not that every social cleavage becomes polarizing in 2022. Most demographic variables remain weak. Nor is the pattern a simple PT/non-PT split. PT evaluation still matters, but the pattern is asymmetric: it produces high between-group divergence without equally high within-group coherence. Ideology is different. It is the only grouping that simultaneously approaches high coherence, high divergence, high non-overlap, and high consensus separation. The 2022 group analysis therefore identifies ideology as the most informative partition among those examined, followed by PT evaluation, while demographic partitions remain weak.

The aggregate contrast with 2006 is that the system becomes more externally separated without

becoming uniformly more internally cohesive. For most groupings, D is higher in 2022 than in 2006, while C is lower or only modestly positive. This means that group consensuses are farther apart, but group members are not necessarily more tightly clustered around those consensuses. Ideology departs from this general pattern: it increases both internal coherence and external separation, which is why it becomes the main group-level signature of the 2022 profile.

Taken together, the global and group-based measures support a more precise interpretation than a simple increase in polarization. The profile becomes more structured over time, but it does not bifurcate into two internally homogeneous and mutually opposed blocs. The global measures show that pairwise antagonism increases from 2006 to 2022. Ψ is already moderate to high in 2006, rises in 2018, and reaches its highest values in 2022, where it remains high even after the candidate set expands beyond the binary Lula–Bolsonaro comparison. Thus, the temporal change is not the emergence of antagonism from a neutral baseline. It is an intensification of dissent from an already nontrivial level.

The reversal-based measures clarify the structure of this intensification. At $m = 2$, the binary comparison mechanically forces all opposition into a single reversal field. This is why $\Psi = R$, $\kappa = 1$, and $ENRP = 1$. Once $m \geq 3$, the structure becomes identifiable: reversal mass can be distributed across multiple opposed rankings. In all three waves, $RHHI$ falls relative to the binary case, and $ENRP > 1$, showing that antagonism does not collapse onto a single divide. But the decline is not the same across years. In 2022, R remains high even for larger candidate sets, $RHHI$ remains above the earlier waves at comparable values of m, and the effective number of reversal pairs is lower. With four alternatives, for example, the effective number of reversal pairs declines from roughly 5.6 in 2006 to 4.0 in 2018 and 3.0 in 2022; with five alternatives, it declines from roughly 18 to 13 and then to about 7. Hence the process is not only one of increasing antagonism. It is also one of increasing compression: more reversal mass is carried by fewer effective regions of the ranking space.

The group-based measures compare how much of this organization is visible under the partitions examined. In 2006, PT evaluation is the only grouping that clearly stands above the demographic partitions. It has higher coherence, divergence, non-overlap, and consensus separation, especially in the binary comparison. But this structure weakens as additional alternatives are admitted. In 2018, the missingness of PT evaluation prevents using it directly, and the relevant political grouping is instead respondents' score for Lula. Lula score is the clearest grouping in that year: it is the only 2018 partition that combines high within-group coherence, high between-group divergence, high non-overlap, and substantial consensus separation. Ideology is also informative, but it remains below Lula score.

By 2022, the ranking of informative groupings changes again. PT evaluation remains informative, but it no longer produces the most internally coherent subprofiles. Its between-group

divergence remains high, which means that PT-defined consensuses are still far apart, but its coherence is lower. In 2022, ideology is the clear outlier. It has the highest coherence, the highest or near-highest divergence, the largest non-overlap, and the largest consensus separation across candidate-set sizes. The resulting sequence is therefore not simply PT versus anti-PT throughout the period. Among the groupings examined, the most informative partition is PT evaluation in 2006, Lula score in 2018, and ideology in 2022.

This matters for the interpretation of the usual "two Brazils" narrative. Binary electoral margins can invite the interpretation that a close race is itself evidence of polarization, because with $m = 2$ the measures collapse to the normalized pair margin and all reversal is forced into one field. But the multiparty profile shows something different. The observed preference profiles are not well described as two impermeable camps with internally unanimous preferences. The evidence points instead to structuration without bifurcation. Dissent becomes stronger, exact reversal mass becomes larger, and the occupied ranking space becomes more compressed, especially in 2022. Yet opposition remains distributed across multiple rankings and multiple reversal fields.

The most accurate summary is therefore not that Brazil simply became polarized, but that the observed preference profiles became more organized. This organization is visible globally as higher pairwise antagonism, higher reversal mass, and greater concentration of reversal fields. At the group level, PT evaluation and Lula score matter a great deal in the observed profiles: PT evaluation is the most informative grouping in 2006, and Lula score is the most informative grouping in 2018. The striking change is that, in 2022, ideology is the strongest group-level organizer among the partitions examined, surpassing PT evaluation across the main measures. This does not make PT or Lula irrelevant to the analysis. Rather, it shows that the strongest observed political grouping changes across waves: from PT evaluation, to Lula score, to ideological self-placement. The observed preference profiles are therefore neither random fragmentation nor pure binary polarization. They form a structured multiparty configuration of overlapping oppositions, increasingly compressed around ideological distinctions in 2022.

Conclusion

In this article, I extend the study of polarization to settings with more than two evaluated objects by focusing on relative evaluations, or preferences, and using the ordinal information contained in survey responses. I develop global and group-based measures of preference-profile structuration and apply them to Brazilian presidential evaluations in 2006, 2018, and 2022. The global measures show increasing antagonism and organization over time, but the magnitude and interpretation of this increase depend on the number of alternatives considered and on whether one measures pairwise opposition, exact reversal mass, or the concentration of reversal fields. Taken together, the results

point to structuration without bifurcation: disagreement becomes stronger and more compressed, but it does not collapse into two internally unanimous and mutually antagonistic camps. The group-based diagnostics show that, among the groupings examined here, the most informative political partition changes across waves: PT evaluation in 2006, Lula score in 2018, and self-identified ideology in 2022. By contrast, most demographic partitions tested here, such as race, sex, age, education, and income, remain weakly informative.

In the context of preferences over candidates, this work drastically qualifies the hegemonic narrative of a divided country and points toward a more fruitful research agenda centered on the dynamic structuring of preference profiles (Perote-Peña and Piggins 2015; Feld and Grofman 1990). More radically, although ordinal information implies a universe of profiles that is formally distinct than that defined by a cardinal interval, the analysis of evaluative social conflicts would be enriched by a pullback to the space of reasons underlying relative evaluations. That is: what is the form taken by the reasons, or rationalizations, offered for such evaluations? In a context of elite radicalization, it is relevant to probe the motivational content of citizens' preferences over candidates, instead of just tallying "affect". The formal framework for this type of investigation into agents' judgment patterns is already well established, but the required data collection would be of a different nature (e.g., focus groups) (List 2011).

References

Ballester, Miguel A. and Guillaume Haeringer (2010). "A Characterization of the Single-Peaked Domain". In: *Social Choice and Welfare* 36.2, pp. 305–322. DOI: 10.1007/s00355-010-0476-3.

Borba, Julian, Ednaldo Ribeiro, and Mario Fuks (2024). "Polarization and Ideology: Exploring the Contextual Nature of Democratic Commitment". In: *Revista de Sociologia e Política* 32.nil, nil. DOI: 10.1590/1678-98732432e006. URL: http://dx.doi.org/10.1590/1678-98732432e006.

Bramson, Aaron, Patrick Grim, Daniel J Singer, et al. (2017). "Understanding polarization: Meanings, measures, and model evaluation". In: *Philosophy of science* 84.1, pp. 115–159.

Bramson, Aaron, Patrick Grim, Daniel J. Singer, et al. (2016). "Disambiguation of Social Polarization Concepts and Measures". In: *The Journal of Mathematical Sociology* 40.2, pp. 80–111. DOI: 10.1080/0022250x.2016.1147443. URL: http://dx.doi.org/10.1080/0022250X.2016.1147443.

Can, Burak, Ali Ihsan Ozkes, and Ton Storcken (2015). "Measuring Polarization in Preferences". In: *Mathematical Social Sciences* 78.nil, pp. 76–79. DOI: 10.1016/j.mathsocsci.2015.09.006. URL: http://dx.doi.org/10.1016/j.mathsocsci.2015.09.006.

DiMaggio, Paul, John Evans, and Bethany Bryson (1996). "Have American's social attitudes become more polarized?" In: *American journal of Sociology* 102.3, pp. 690–755.

Dowding, Keith and Charles Miller (2025). "Explanatory Pluralism in Political Science". In: *Journal of Theoretical Politics* nil.nil, p. 09516298251403391. DOI: 10.1177/09516298251403391. URL: http://dx.doi.org/10.1177/09516298251403391.

Elkind, Edith, Martin Lackner, and Dominik Peters (2022). "Preference restrictions in computational social choice: A survey". In: *arXiv preprint*. arXiv: 2205.09092 [cs.GT]. URL: https://arxiv.org/abs/2205.09092.

Elsner, Mark, Grace Atkinson, and Saadia Zahidi (Jan. 2025). *Global Risks Report 2025*. Tech. rep. Geneva: World Economic Forum. URL: https://www.weforum.org/publications/global-risks-report-2025/digest/.

Esteban, Joan-Maria and Debraj Ray (1994). "On the measurement of polarization". In: *Econometrica: Journal of the Econometric Society*, pp. 819–851.

Feld, Scott L and Bernard Grofman (1988). "Ideological consistency as a collective phenomenon". In: *American Political Science Review* 82.3, pp. 773–788.

— (1990). "Collectivities as actors: Consistency of collective choices". In: *Rationality and Society* 2.4, pp. 429–448.

Fiorina, Morris P. and Samuel J. Abrams (2008). "Political Polarization in the American Public". In: *Annual Review of Political Science* 11.1, pp. 563–588. DOI: 10.1146/annurev.polisci.11.053106.153836. URL: http://dx.doi.org/10.1146/annurev.polisci.11.053106.153836.

Flache, Andreas et al. (2017). "Models of Social Influence: Towards the Next Frontiers". In: *Journal of Artificial Societies and Social Simulation* 20.4, p. 2. ISSN: 1460-7425. DOI: 10.18564/jasss.3521. URL: http://jasss.soc.surrey.ac.uk/20/4/2.html.

Fuks, Mario and Pedro Henrique Marques (2022). "Polarização E Contexto: Medindo E Explicando a polarização política No Brasil". In: *Opinião Pública* 28.3, pp. 560–593. DOI: 10.1590/1807-01912022283560. URL: http://dx.doi.org/10.1590/1807-01912022283560.

Gerring, John (2012). "Mere Description". In: *British Journal of Political Science* 42.4, pp. 721–746. DOI: 10.1017/s0007123412000130.

Holbrook, Allyson L and James N Druckman (2011). "Attitude change experiments in political science". In: *Cambridge handbook of experimental political science* 1, pp. 141–154.

Igersheim, Herrade et al. (2022). "Comparing Voting Methods: 2016 Us Presidential Election". In: *European Journal of Political Economy* 71.nil, p. 102057. DOI: 10.1016/j.ejpoleco.2021.102057.

Iyengar, Shanto, Gaurav Sood, and Yphtach Lelkes (2012). "Affect, Not Ideology". In: *Public Opinion Quarterly* 76.3, pp. 405–431. DOI: 10.1093/poq/nfs038. URL: http://dx.doi.org/10.1093/poq/nfs038.

Kaminski, Marek M. (2015). "Empirical examples of voting paradoxes". In: *Handbook of social choice and voting*. Edward Elgar Publishing.

Kekkonen, Arto and Tuomas Ylä-Anttila (2021). "Affective Blocs: Understanding Affective Polarization in Multiparty Systems". In: *Electoral Studies* 72.nil, p. 102367. DOI: 10.1016/j.electstud.2021.102367. URL: http://dx.doi.org/10.1016/j.electstud.2021.102367.

Lachat, Romain and Jean-François Laslier (2024). "Alternatives To Plurality Rule for Single-Winner Elections: When Do They Make a Difference?" In: *European Journal of Political Economy* 81.nil, p. 102505. DOI: 10.1016/j.ejpoleco.2024.102505. URL: http://dx.doi.org/10.1016/j.ejpoleco.2024.102505.

List, Christian (2011). "The Theory of Judgment Aggregation: an Introductory Review". In: *Synthese* 187.1, pp. 179–207. DOI: 10.1007/s11229-011-0025-3. URL: http://dx.doi.org/10.1007/s11229-011-0025-3.

Maciel, Marcelo V. (2024). "Was Bolsonaro's 2018 electoral victory an institutional accident?" In: *European Journal of Political Economy* 83, p. 102548.

Maciel, Marcelo V. and André C.R. Martins (2020). "Ideologically Motivated Biases in a Multiple Issues Opinion Model". In: *Physica A: Statistical Mechanics and its Applications* 553.nil, p. 124293. DOI: 10.1016/j.physa.2020.124293.

Mason, Lilliana (2015). ""I disrespectfully agree": The differential effects of partisan sorting on social and issue polarization". In: *American journal of political science* 59.1, pp. 128–145.

Mehlhaff, Isaac D. (2023). "A Group-Based Approach To Measuring Polarization". In: *American Political Science Review* 118.3, pp. 1518–1526. DOI: 10.1017/s0003055423001041. URL: http://dx.doi.org/10.1017/S0003055423001041.

Mouw, Ted and Michael E Sobel (2001). "Culture wars and opinion polarization: the case of abortion". In: *American Journal of Sociology* 106.4, pp. 913–943.

Narens, Louis (2007). *Introduction to the Theories of Measurement and Meaningfulness and the Use of Symmetry in Science*. Psychology Press.

Narens, Louis and Brian Skyrms (2020). *The Pursuit of Happiness: Philosophical and Psychological Foundations of Utility*. Oxford University Press.

Nunes, Felipe and Thomas Traumann (2023). *Biografia do abismo*. HARLEQUIN.

Patty, John W. and Elizabeth Maggie Penn (2014). "Analyzing Big Data: Social Choice and Measurement". In: *PS: Political Science & Politics* 48.01, pp. 95–101. DOI: 10.1017/s1049096514001814.

Patty, John W. and Elizabeth Maggie Penn (2019). "Measuring fairness, inequality, and big data: Social choice since Arrow". In: *Annual Review of Political Science* 22, pp. 435–460.

Perote-Peña, Juan and Ashley Piggins (2015). "A Model of Deliberative and Aggregative Democracy". In: *Economics and Philosophy* 31.1, pp. 93–121. DOI: 10.1017/s0266267114000418.

Popov, Sergey V, Anna Popova, and Michel Regenwetter (2014). "Consensus in organizations: Hunting for the social choice conundrum in APA elections." In: *Decision* 1.2, p. 123.

Regenwetter, Michel et al. (2006). *Behavioral social choice: probabilistic models, statistical inference, and applications*. Cambridge University Press.

Rothers, Adrian (2025). "The Poles in Polarization: Social Categorization and Affective Polarization in Multiparty Systems". In: *Electoral Studies* 95.nil, p. 102908. DOI: 10.1016/j.electstud.2025.102908. URL: http://dx.doi.org/10.1016/j.electstud.2025.102908.

Samuels, David, Fernando Mello, and Cesar Zucco (2023). "Partisan Stereotyping and Polarization in Brazil". In: *Latin American Politics and Society* 66.2, pp. 47–71. DOI: 10.1017/lap.2023.38. URL: http://dx.doi.org/10.1017/lap.2023.38.

Samuels, David J and Cesar Zucco (2018). *Partisans, antipartisans, and nonpartisans: voting behavior in Brazil*. Cambridge University Press.

Sen, Amartya (2007). *Identity and violence: The illusion of destiny (issues of our time)*. WW Norton & Company.

— (2018). *Collective choice and social welfare*. Harvard University Press.

Shugart, Matthew and Rein Taagepera (2017). *Votes from seats: Logical models of electoral systems*. Cambridge University Press.

Szufa, Stanislaw Andrzej (2024). "Map of Elections". Ph.D. thesis. Krakow, Poland: AGH University, Faculty of Computer Science.

Zhang, Zhongheng (2016). "Multiple imputation with multivariate imputation by chained equation (MICE) package". In: *Annals of translational medicine* 4.2, p. 30.

Zwicker, William S (2016). "Introduction to the Theory of Voting." In: *Handbook of computational social choice*. Ed. by Felix Brandt et al.

A Mathematical Appendix

A.1 Decomposition of D into within-group dispersion and consensus separation

This appendix shows why the group divergence statistic D, under the member-to-other-consensus definition used in this paper, can be decomposed into a within-group dispersion component and a consensus-separation component. The result depends on defining D as the average distance from

members of one group to the consensus rankings of other groups. It does not apply to a statistic defined only as the distance between group consensuses.

Let A be the active candidate set, with $|A| = m$, and let $\mathcal{L}(A)$ denote the set of linear orders over A. For each group G_g of size n_g, define the normalized average Kendall loss of a candidate consensus ranking $\rho \in \mathcal{L}(A)$ as

$$\ell_g(\rho) = \frac{1}{n_g} \sum_{x \in G_g} \frac{d_\tau(x, \rho)}{\binom{m}{2}},$$

where x is the observed ranking of an individual member of G_g, and d_τ is Kendall tau distance.

The consensus set of group g is the set of rankings that minimize this loss:

$$M_g = \arg\min_{\rho \in \mathcal{L}(A)} \ell_g(\rho).$$

When M_g is not a singleton, consensus-dependent quantities are averaged over the minimizing rankings. Let $\bar{\ell}_g(M_h)$ denote the average loss of group g's members relative to the consensus set of group h:

$$\bar{\ell}_g(M_h) = \frac{1}{|M_h|} \sum_{\rho \in M_h} \ell_g(\rho).$$

The within-group dispersion of G_g around its own consensus set is its minimum average loss:

$$W_g = \bar{\ell}_g(M_g) = \min_{\rho \in \mathcal{L}(A)} \ell_g(\rho).$$

Because every ranking $\rho \in M_g$ achieves this same minimum, averaging over M_g does not change the value.

The raw coherence of group g is

$$C_g^\star = 1 - W_g.$$

Aggregating by group size gives

$$C^\star = \sum_{g=1}^{k} \frac{n_g}{n} C_g^\star, \qquad n = \sum_{g=1}^{k} n_g.$$

Therefore the aggregate within-group dispersion is

$$W = \sum_{g=1}^{k} \frac{n_g}{n} W_g = 1 - C^\star.$$

The raw coherence statistic $C^\star$ is not bounded below by 0, but by $1/2$. To see this, fix any

ranking $\rho \in \mathcal{L}(A)$, and let ρ^{rev} be its reverse. For any individual ranking x,

$$d_\tau(x, \rho) + d_\tau(x, \rho^{\text{rev}}) = \binom{m}{2}.$$

After normalizing and averaging over members of group g, this implies

$$\ell_g(\rho) + \ell_g(\rho^{\text{rev}}) = 1.$$

Hence at least one of ρ or ρ^{rev} has average normalized loss no greater than $1/2$. Since W_g is the minimum of $\ell_g(\rho)$ over all rankings,

$$W_g \leq \frac{1}{2}.$$

It follows that

$$C_g^\star = 1 - W_g \geq \frac{1}{2},$$

and therefore $C^\star \geq 1/2$. This motivates the rebased coherence statistic

$$C = 2C^\star - 1,$$

which places the no-structure baseline at zero.

The group divergence statistic is defined as the average distance from members of one group to the consensus set of other groups:

$$D = \frac{1}{k-1} \sum_{g=1}^{k} \sum_{h \neq g} \frac{n_g}{n} \bar{\ell}_g(M_h).$$

For each ordered pair of distinct groups (g, h), add and subtract group g's own internal dispersion W_g:

$$\bar{\ell}_g(M_h) = W_g + \left[\bar{\ell}_g(M_h) - W_g\right].$$

Substituting into the definition of D yields

$$D = \underbrace{\frac{1}{k-1} \sum_{g=1}^{k} \sum_{h \neq g} \frac{n_g}{n} W_g}_{\text{within-group dispersion}} + \underbrace{\frac{1}{k-1} \sum_{g=1}^{k} \sum_{h \neq g} \frac{n_g}{n} \left[\bar{\ell}_g(M_h) - W_g\right]}_{\text{consensus separation}}.$$

The first term simplifies because W_g depends only on g, not on h. For each fixed group g, there

are $k-1$ groups $h \neq g$. Hence

$$\frac{1}{k-1}\sum_{g=1}^{k}\sum_{h\neq g}\frac{n_g}{n}W_g = \frac{1}{k-1}\sum_{g=1}^{k}(k-1)\frac{n_g}{n}W_g = \sum_{g=1}^{k}\frac{n_g}{n}W_g = W.$$

Since $W = 1 - C^\star$, the decomposition becomes

$$D = (1 - C^\star) + S,$$

where S is the consensus-separation statistic:

$$S = \frac{1}{k-1}\sum_{g=1}^{k}\sum_{h\neq g}\frac{n_g}{n}\left[\bar{\ell}_g(M_h) - W_g\right].$$

Equivalently,

$$S = D - (1 - C^\star).$$

If the rebased coherence statistic $C = 2C^\star - 1$ is used, then

$$C^\star = \frac{C+1}{2},$$

so

$$1 - C^\star = \frac{1-C}{2}.$$

Thus the same decomposition can be written on the rebased scale as

$$D = \frac{1-C}{2} + S,$$

or equivalently,

$$S = D - \frac{1-C}{2}.$$

The consensus-separation statistic S is non-negative. By definition,

$$W_g = \min_{\rho\in\mathcal{L}(A)} \ell_g(\rho).$$

Therefore, for any ranking $\rho \in \mathcal{L}(A)$,

$$\ell_g(\rho) \geq W_g.$$

In particular, for every ranking $\rho \in M_h$,

$$\ell_g(\rho) \geq W_g.$$

Averaging over all rankings in M_h preserves the inequality:

$$\bar{\ell}_g(M_h) \geq W_g.$$

Hence each bracketed term in S is non-negative:

$$\bar{\ell}_g(M_h) - W_g \geq 0.$$

Since S is a weighted average of non-negative terms,

$$S \geq 0.$$

The statistic S equals zero when the consensus set of every other group is also optimal, on average, for the group being evaluated. It becomes positive when other groups' consensus rankings are worse summaries of group g's members than group g's own consensus set. Thus S should be interpreted as consensus separation: the part of D that remains after removing the within-group dispersion already built into distances from individual rankings to consensus rankings.

A.2 On RHHI and e(m) profiles

For the profile $e(m)$, each of the $m!$ possible orderings occurs with frequency $1/m!$. Consider a pair of reversal orderings $(r, -r)$. Since both have the same frequency, the local reversal component is $\frac{2}{m!}$. There are $m!/2$ such pairs, therefore $R\big(e(m)\big) = \frac{m!}{2}\,\frac{2}{m!} = 1$. Note that $q_i = 2/m!$. Applying the Herfindahl–Hirschman Index to $\mathbf{q} = (q_1, \ldots, q_{m!/2})$, we obtain $\kappa\big(q(e(m))\big) = \frac{m!}{2}\left(\frac{2}{m!}\right)^2 = \frac{2}{m!}$. Finally, the reversal-based polarization measure is the geometric mean between intensity (R) and its concentration (κ):

$$\mathrm{RHHI}\big(e(m)\big) = \sqrt{R\big(e(m)\big)\ \kappa\big(\mathrm{q}(e(m))\big)} = \sqrt{\frac{2}{m!}}\,.$$

A.3 Derivations for m=2

For $\kappa = 1$ when $m = 2$ note that the number of possible rankings is $2! = 2$. The set of rankings is $\{A \succ B,\ B \succ A\}$, and there is only one possible reversed pair: $(A \succ B,\ B \succ A)$. Let p and $1 - p$ be the proportions of voters with each ranking. Then, the local reversal component is

$r = 2 \cdot \min(p, 1-p)$. Since there is only this pair, the total reversal is just r. Thus, the normalized distribution has $q_1 = r/R = 1$, which implies $\kappa = \sum_i q_i^2 = 1^2 = 1$.

Now, for $m = 2$ alternatives, Ψ and R coincide. Let there be $m = 2$ alternatives, A and B, and n voters. Denote

$$n_{AB} = \#\{i : A \succ_i B\}, \qquad n_{BA} = n - n_{AB}.$$

By Can, Ozkes, and Storcken (2015), for $m = 2$ there is only one unordered pair $\{A, B\}$, so

$$\Psi(p) = \frac{n - d_{AB}(p)}{n\binom{2}{2}} = \frac{n - d_{AB}(p)}{n},$$

where

$$d_{AB}(p) = \left|n_{AB} - n_{BA}\right| = \left|n_{AB} - (n - n_{AB})\right| = \left|2n_{AB} - n\right|.$$

Hence

$$\Psi(p) = 1 - \frac{|2n_{AB} - n|}{n}.$$

There are only two linear orders, AB and BA, which are reversals. Let the proportions be

$$p(AB) = \frac{n_{AB}}{n}, \qquad p(BA) = \frac{n_{BA}}{n}.$$

By definition,

$$R(p) = 2\min\{p(AB), p(BA)\} = \frac{2}{n}\min\{n_{AB},\, n - n_{AB}\}.$$

Using $\min\{x, y\} = \frac{x+y-|x-y|}{2}$ with $x = n_{AB}$, $y = n - n_{AB}$, we obtain

$$\min\{n_{AB}, n - n_{AB}\} = \frac{n - |n - 2n_{AB}|}{2},$$

and

$$R(p) = \frac{2}{n} \cdot \frac{n - |n - 2n_{AB}|}{2} = 1 - \frac{|n - 2n_{AB}|}{n} = 1 - \frac{|2n_{AB} - n|}{n}.$$

Thus, we have $\Psi(p) = R(p)$ for $m = 2$.

B Methodological Appendix

B.1 Pattern-conditional linearization with Jeffreys smoothing

The survey records evaluations as ratings rather than complete strict rankings. For a respondent i, let A_m denote the candidate set and let $x_i(a)$ be the imputed rating assigned to candidate $a \in A_m$. These ratings induce a weak order w_i over A_m. Equivalently, A_m is partitioned into indifference

classes

$$B_{i1}, B_{i2}, \ldots, B_{iK_i},$$

where all candidates in the same class receive the same rating and the classes are ordered from higher to lower rating:

$$x_i(a) > x_i(b) \quad \text{whenever} \quad a \in B_{ik},\ b \in B_{i\ell},\ k < \ell.$$

Linearization preserves this between-class ordering and resolves only within-class indifferences.

Let $\mathcal{E}(w)$ be the set of all strict linear extensions compatible with weak order w. Thus,

$$\mathcal{E}(w) = \{\pi \in \mathcal{L}(A_m) : \pi \text{ preserves the strict comparisons in } w\},$$

where $\mathcal{L}(A_m)$ is the set of all $m!$ linear orders over A_m. If the weak order has indifference classes $B_1, \ldots, B_K$, then

$$|\mathcal{E}(w)| = \prod_{k=1}^{K} |B_k|!.$$

The random-tie linearizer samples uniformly from $\mathcal{E}(w)$:

$$p_{\text{unif}}(\pi \mid w) = \frac{1}{|\mathcal{E}(w)|}, \qquad \pi \in \mathcal{E}(w).$$

The pattern-conditional linearizer instead estimates a conditional distribution over compatible refinements. Let $n_w(\pi)$ denote the number of observed cases in the relevant empirical stratum with weak-order pattern w whose strict refinement is $\pi \in \mathcal{E}(w)$. The raw empirical estimator would be

$$\widehat{p}(\pi \mid w) = \frac{n_w(\pi)}{\sum_{\pi' \in \mathcal{E}(w)} n_w(\pi')}.$$

This estimator is undesirable in sparse strata because it assigns probability zero to unobserved refinements and probability one to a refinement observed only once when no alternatives have been observed.

I therefore use Jeffreys smoothing. For each compatible refinement $\pi \in \mathcal{E}(w)$, define

$$\widetilde{p}_\alpha(\pi \mid w) = \frac{n_w(\pi) + \alpha}{\sum_{\pi' \in \mathcal{E}(w)} \{n_w(\pi') + \alpha\}}, \qquad \alpha = \frac{1}{2}.$$

Equivalently, if

$$N_w = \sum_{\pi' \in \mathcal{E}(w)} n_w(\pi'),$$

then

$$\widetilde{p}_{1/2}(\pi \mid w) = \frac{n_w(\pi) + 1/2}{N_w + \frac{1}{2}|\mathcal{E}(w)|}.$$

This is the posterior mean under a symmetric Dirichlet prior with parameter $\alpha = 1/2$ over the vector of refinement probabilities. It is symmetric across refinements, prevents degenerate zero and one probabilities in sparse cells, and regularizes less strongly than Laplace smoothing ($\alpha = 1$).

The behavior of the estimator is straightforward. If no empirical refinements are observed for a weak-order pattern, $N_w = 0$, then

$$\widetilde{p}_{1/2}(\pi \mid w) = \frac{1}{|\mathcal{E}(w)|},$$

so the procedure reduces to uniform tie-breaking. If N_w is large, the pseudo-counts become negligible and

$$\widetilde{p}_{1/2}(\pi \mid w) \approx \widehat{p}(\pi \mid w).$$

Thus the rule interpolates between uniform tie-breaking in empty or very sparse strata and empirical pattern-conditional tie-breaking in well-populated strata.

The linearization step for respondent i is therefore:

$$\pi_i \sim \widetilde{p}_{1/2}(\cdot \mid w_i), \qquad \pi_i \in \mathcal{E}(w_i).$$

This preserves the ordinal information contained in the ratings while allowing recurrent empirical refinements of ties to inform the completed ranking. The purpose is not to recover a latent "true" strict order behind every tie, but to avoid injecting purely uniform noise when the data contain stable tie-breaking structure.

B.2 Nested variance decomposition

For a fixed analysis cell, defined by wave, scenario, candidate-set size m, measure, grouping, and pipeline specification, let

$$Y_{brk}$$

denote the value of the measure computed from bootstrap replicate $b \in \{1, \ldots, B\}$, imputation replicate $r \in \{1, \ldots, R\}$, and linearization replicate $k \in \{1, \ldots, K\}$. The indices are nested: linearization varies within imputed datasets, imputation varies within bootstrap pseudo-profiles, and bootstrap resampling varies over pseudo-profiles.

Define the lower-level conditional means

$$\bar{Y}_{br\cdot} = \frac{1}{K}\sum_{k=1}^{K} Y_{brk},$$

and

$$\bar{Y}_{b\cdot\cdot} = \frac{1}{R}\sum_{r=1}^{R} \bar{Y}_{br\cdot}.$$

The overall mean for the fixed analysis cell is

$$\bar{Y}_{\cdots} = \frac{1}{B}\sum_{b=1}^{B} \bar{Y}_{b\cdots}.$$

The bootstrap component is the variance across bootstrap-level conditional means:

$$V_{\text{boot}} = \text{Var}_b(\bar{Y}_{b\cdot\cdot}).$$

This term captures variation induced by changing the weighted pseudo-profile while averaging over the imputation and linearization replicates nested within that pseudo-profile.

The imputation component is the average, over bootstrap replicates, of the within-bootstrap variance across imputation-level conditional means:

$$V_{\text{imp}} = \frac{1}{B}\sum_{b=1}^{B} \text{Var}_r(\bar{Y}_{br\cdot}).$$

This term captures variation induced by different imputations of the same bootstrap pseudo-profile, after averaging over the linearization replicates nested within each imputed dataset.

The linearization component is the average, over bootstrap and imputation replicates, of the within-(b, r) variance across linearizations:

$$V_{\text{lin}} = \frac{1}{BR}\sum_{b=1}^{B}\sum_{r=1}^{R} \text{Var}_k(Y_{brk}).$$

This term captures variation induced by completing weak preferences into linear orders conditional on a fixed bootstrap pseudo-profile and a fixed imputed dataset.

The reported total is the sum of these nested components:

$$V_{\text{tot}} = V_{\text{boot}} + V_{\text{imp}} + V_{\text{lin}}.$$

With the balanced-tree and population-variance convention used in the replication code, this total equals the empirical variance of the leaf-level values:

$$V_{\text{tot}} = \text{Var}_{b,r,k}(Y_{brk}).$$

The point of the decomposition is therefore not to define a different total variance. It is to partition the empirical simulation variance according to the nested stochastic operations that generated it. Variation across k is interpreted as linearization variability only after conditioning on b and r; variation across r is interpreted as imputation variability only after conditioning on b; and variation across b is interpreted as bootstrap variability only after averaging over the lower-level stochastic operations.

Explicitly, the implementation uses the population-variance convention

$$\text{Var}_b(\bar{Y}_{b\cdot\cdot}) = \frac{1}{B}\sum_{b=1}^{B}\left(\bar{Y}_{b\cdot\cdot} - \bar{Y}_{\cdot\cdot\cdot}\right)^2,$$

with analogous formulas for Var_r and Var_k. Under this convention, a component with only one realized branch is set to zero. Thus, if $B = 1$, the bootstrap component is zero; if $R = 1$, the imputation component is zero; and if $K = 1$, the linearization component is zero.

The decomposition is computed within fixed analysis cells. In particular, candidate-set size m is part of the conditioning information, not a stochastic source of variation. This is important because changing m changes the ranking space, the number of possible linear orders, the sparsity of the profile, and the scale on which several measures vary. Comparisons across pipeline specifications are therefore made within the same wave, scenario, candidate-set size m, measure, and grouping. Summaries that pool over multiple values of m are used only as compact descriptive diagnostics across analysis cells, not as the primary variance decomposition.

The decomposition is descriptive. It does not introduce an additional population model for the measures. Its purpose is to report how much of the simulation variability in the measured profile, within a fixed analysis cell, is attributable to weighted resampling, missing-data imputation, and stochastic linearization.

A compact visualization of this decomposition is shown in Figure 6. The figure reports, for the 2022 wave and the selected scenario, the distribution of the three variance components across pipeline specifications, separately for each measure and candidate-set size m. The horizontal axis gives m, and within each value of m the boxplots compare the bootstrap, imputation, and linearization components. For the group-based measures, the boxplots pool the corresponding analysis cells across the available groupings, so the figure should be read as a descriptive summary of how the decomposition behaves across the set of group diagnostics, not as a statement about one

particular grouping variable.

Several features stand out. First, the variance values are generally very small in absolute magnitude. In most panels the components lie close to zero, often in the range of small fractions of a percentage point. This is itself a substantive result: although the pipeline contains several stochastic steps, the resulting measures are typically quite stable within a fixed analysis cell. The decomposition is therefore not uncovering large instabilities in the measurement procedure, but rather a modest amount of simulation variability whose relative composition can still be meaningfully compared across measures.

Second, most of the variance comes from the bootstrap stage. Across nearly all measures and values of m, the bootstrap component is visibly larger than the imputation and linearization components. This means that most of the simulation variability comes from changing the weighted pseudo-profile rather than from the downstream stochastic operations applied to that pseudo-profile. This pattern is especially clear for Ψ, R, HHI, and $RHHI$, but it also holds for the group-based measures C, D, S, and $1-O$. In other words, once a bootstrap pseudo-profile is fixed, the additional variability introduced by missing-data imputation and by stochastic linearization is usually much smaller than the variability already induced by resampling.

Third, the imputation and linearization components are typically small and of similar order of magnitude, although their relative importance varies somewhat across measures. Imputation is sometimes more visible for measures such as R, $RHHI$, and $1-O$, especially at larger values of m, while linearization occasionally becomes more salient for some group-based measures. Even so, both components remain well below the bootstrap contribution in most cases.

Fourth, there is some heterogeneity across measures and across m. Measures such as D and S occasionally exhibit somewhat larger bootstrap dispersion, and HHI and $RHHI$ show more uneven behavior as m changes. This is unsurprising. Different measures respond differently to the geometry and sparsity of the profile, and increasing m changes the ranking space itself. Still, the broad qualitative pattern remains the same: candidate-set size matters for the level of the components, but it does not overturn the basic conclusion that the decomposition is dominated by bootstrap variability.

The corresponding figures for the other years are very similar. The exact magnitudes differ somewhat across waves and scenarios, but the same basic structure reappears: total simulation variability is generally small, most of it comes from bootstrap resampling, and the imputation and linearization components are comparatively modest. The decomposition therefore supports the view that the primary source of uncertainty in these estimates is the sampling variation represented by the weighted bootstrap, rather than instability in the subsequent handling of missing values or weak-order linearization.

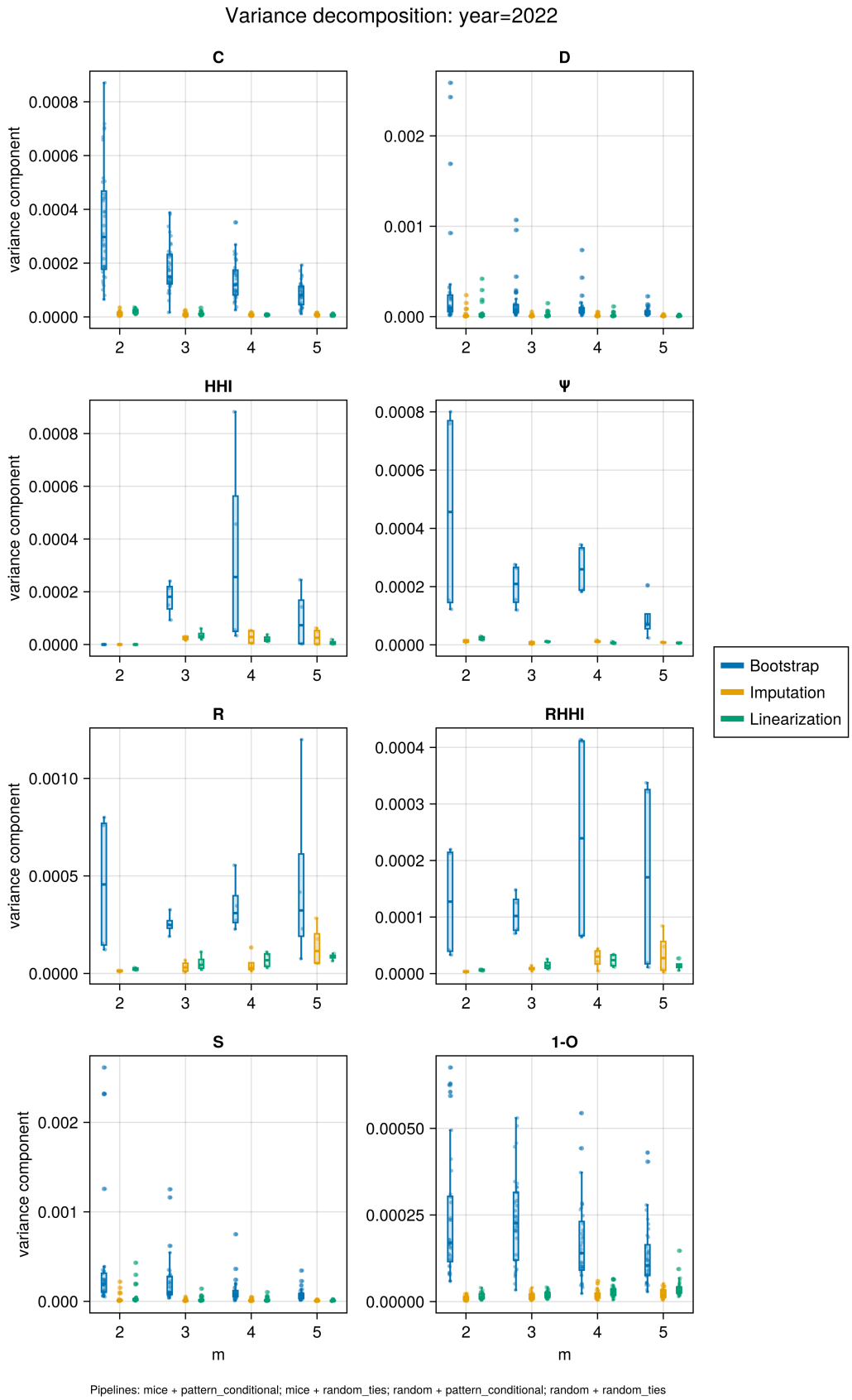


Figure 6: Nested variance decomposition by measure and candidate-set size, 2022. For the group-based measures, the distributions pool the relevant analysis cells across available groupings, so the figure is intended as a compact descriptive summary of the sources of simulation variability.

B.3 Candidate Set Selection

For 2006, the availability criterion and substantive relevance coincide closely. The five most available figures are Lula, Geraldo Alckmin, Heloísa Helena, José Serra, and Cristóvam Buarque.[20] This set captures the incumbent president, the main opposition candidate, a relevant left-wing dissident candidacy, and nationally salient presidential or presidentially relevant figures. I therefore use this order as the 2006 candidate sequence.

The 2018 case requires a different decision rule. If alternatives are selected only by weighted missingness, the first names are Lula, Dilma Rousseff, Michel Temer, Jair Bolsonaro, Fernando Haddad, Marina Silva, Aécio Neves, Ciro Gomes, and Geraldo Alckmin.[21] This ordering is informative, but it is not an adequate rule for constructing the 2018 preference profile. It selects highly recognizable political figures, including Lula, Dilma Rousseff, and Michel Temer, rather than only electorally relevant presidential alternatives. Since Lula did not appear on the 2018 ballot, I do not include him as an alternative in the 2018 preference profile. Instead, I use respondents' evaluation of Lula as a political-evaluative partition in the group analysis. The 2018 candidate sequence is therefore Fernando Haddad, Jair Bolsonaro, Ciro Gomes, Geraldo Alckmin, and Marina Silva. This set captures the principal electoral alternatives in the presidential contest while avoiding the conflation of notoriety with candidacy.

For 2022, the availability criterion again aligns more closely with electoral relevance, with one important exception. The lowest-missingness figures are Lula, Bolsonaro, Ciro Gomes, Marina Silva, Geraldo Alckmin, and Simone Tebet.[22] However, Geraldo Alckmin was Lula's vice-presidential running mate in 2022 rather than an independent presidential alternative. For this reason, I replace Alckmin with Simone Tebet, who was an actual presidential candidate and occupied a substantively relevant position in the electoral field. The 2022 candidate sequence is thus Lula, Bolsonaro, Ciro Gomes, Simone Tebet, and Marina Silva.

Table 4: Effective ranking and reversal-pair evolution

m	EO 2006	ENRP 2006	EO 2018	ENRP 2018	EO 2022	ENRP 2022
2	1.78	1.00	1.86	1.00	1.99	1.00
3	3.84	1.96	3.79	2.18	3.00	1.49
4	11.08	5.53	9.49	4.08	6.41	2.98
5	34.82	18.03	32.33	13.73	20.21	7.26

Table 5: Aggregate separation ratio Λ for 2006 / main_2006.

m	Age	Education	Ideology	Income	PT	Sex
2	1.000	1.210	1.000	1.277	1.983	1.000
3	1.000	1.116	1.000	1.247	1.340	1.000
4	1.021	1.114	1.026	1.191	1.331	1.000
5	1.009	1.138	1.017	1.167	1.185	1.000

Λ is the aggregate separation ratio $\Lambda = (\sum_g \pi_g D_g)/(\sum_g \pi_g W_g)$. Values equal to 1 indicate no excess separation; values above 1 indicate that outgroup consensuses are farther away than the ingroup consensus, relative to within-group dispersion.

Table 6: Aggregate separation ratio Λ for 2018 / main_2018.

m	Age	Education	Ideology	Income	LulaScoreGroup	Race	Religion	Sex
2	1.129	1.074	1.427	1.045	2.325	1.000	1.229	1.000
3	1.088	1.103	1.312	1.055	1.988	1.060	1.183	1.000
4	1.083	1.065	1.256	1.080	1.837	1.052	1.176	1.027
5	1.047	1.066	1.216	1.093	1.649	1.051	1.157	1.042

Λ is the aggregate separation ratio $\Lambda = (\sum_g \pi_g D_g)/(\sum_g \pi_g W_g)$. Values equal to 1 indicate no excess separation; values above 1 indicate that outgroup consensuses are farther away than the ingroup consensus, relative to within-group dispersion.

Table 7: Aggregate separation ratio Λ for 2022 / main_2022.

m	Abortion	Age	Education	Ideology	Income	PT	Race	Religion	Sex
2	1.222	1.000	1.112	2.061	1.161	1.441	1.048	1.306	1.067
3	1.177	1.032	1.128	1.909	1.153	1.441	1.064	1.275	1.082
4	1.156	1.035	1.130	1.725	1.153	1.335	1.058	1.224	1.122
5	1.136	1.022	1.111	1.551	1.135	1.281	1.067	1.204	1.133

Λ is the aggregate separation ratio $\Lambda = (\sum_g \pi_g D_g)/(\sum_g \pi_g W_g)$. Values equal to 1 indicate no excess separation; values above 1 indicate that outgroup consensuses are farther away than the ingroup consensus, relative to within-group dispersion.

C Supporting Results

C.1 Effective Rankings

C.2 Interpreting S

The appendix tables report the aggregate separation ratio, Λ, for each main scenario. Λ compares the average distance from respondents to outgroup consensuses with their average distance to the ingroup consensus. A value of $\Lambda = 1$ indicates no excess separation: outgroup consensuses are no farther away than the ingroup consensus. Values above 1 indicate increasing relative separation. For example, $\Lambda = 1.5$ means that outgroup consensuses are 50 percent farther away than the ingroup consensus, while $\Lambda = 2$ means they are twice as far away.

Across the three main scenarios, the largest values of Λ are associated with political groupings rather than standard demographic groupings. In 2006, partisan identification with PT is the clearest source of separation, especially for smaller candidate sets. For m = 2, the PT grouping has $\Lambda = 1.983$, indicating that outgroup consensuses are almost twice as far away as the ingroup consensus. This contrast weakens as the candidate set expands, falling to $\Lambda = 1.185$ by m = 5.

In 2018, LulaScoreGroup is by far the strongest grouping. Its separation ratio is $\Lambda = 2.325$ for m = 2, meaning that outgroup consensuses are more than twice as far away as the ingroup consensus. The ratio declines monotonically as m increases, but remains substantively large even at m = 5, where $\Lambda = 1.649$. Ideology is the second most separated grouping in 2018, with values between 1.216 and 1.427, while most demographic groupings remain closer to 1.

In 2022, ideology becomes the dominant source of separation. The ideological grouping reaches $\Lambda = 2.061$ for m = 2, again indicating that outgroup consensuses are roughly twice as far away as the ingroup consensus. The ratio decreases as additional candidates enter the ranking, but remains high at $\Lambda = 1.551$ for m = 5. PT is consistently the next strongest grouping, followed by religion, while age, race, sex, income, and education generally show smaller ratios.

Overall, the tables show that the strongest relative separation is concentrated in politically meaningful groupings: PT in 2006, evaluations of Lula in 2018, and ideology in 2022. Standard demographic groupings often produce ratios close to 1, suggesting limited excess separation relative to within-group dispersion. The decline in Λ as m increases indicates that the most compressed candidate sets tend to reveal the sharpest group separation, while larger candidate sets introduce ad-

[20]The weighted missingness order for 2006 is: Lula, 1.91%; Geraldo Alckmin, 5.68%; Heloísa Helena, 8.10%; José Serra, 8.71%; and Cristóvam Buarque, 19.18%.

[21]The weighted missingness order for the leading 2018 figures is: Lula, 2.79%; Dilma Rousseff, 4.15%; Michel Temer, 5.95%; Jair Bolsonaro, 7.18%; Fernando Haddad, 9.62%; Marina Silva, 11.49%; Aécio Neves, 13.65%; Ciro Gomes, 14.21%; and Geraldo Alckmin, 14.68%.

[22]The weighted missingness order for the leading 2022 figures is: Lula, 1.43%; Bolsonaro, 1.43%; Ciro Gomes, 8.12%; Marina Silva, 11.14%; Geraldo Alckmin, 11.30%; and Simone Tebet, 18.29%.

ditional within-group heterogeneity and reduce the relative distance between ingroup and outgroup consensuses.